Digital Image Processing

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Lecture 10

Image Segmentation - II

Overview

1 Recap of Traditional Methods

2 Evaluation Metrics

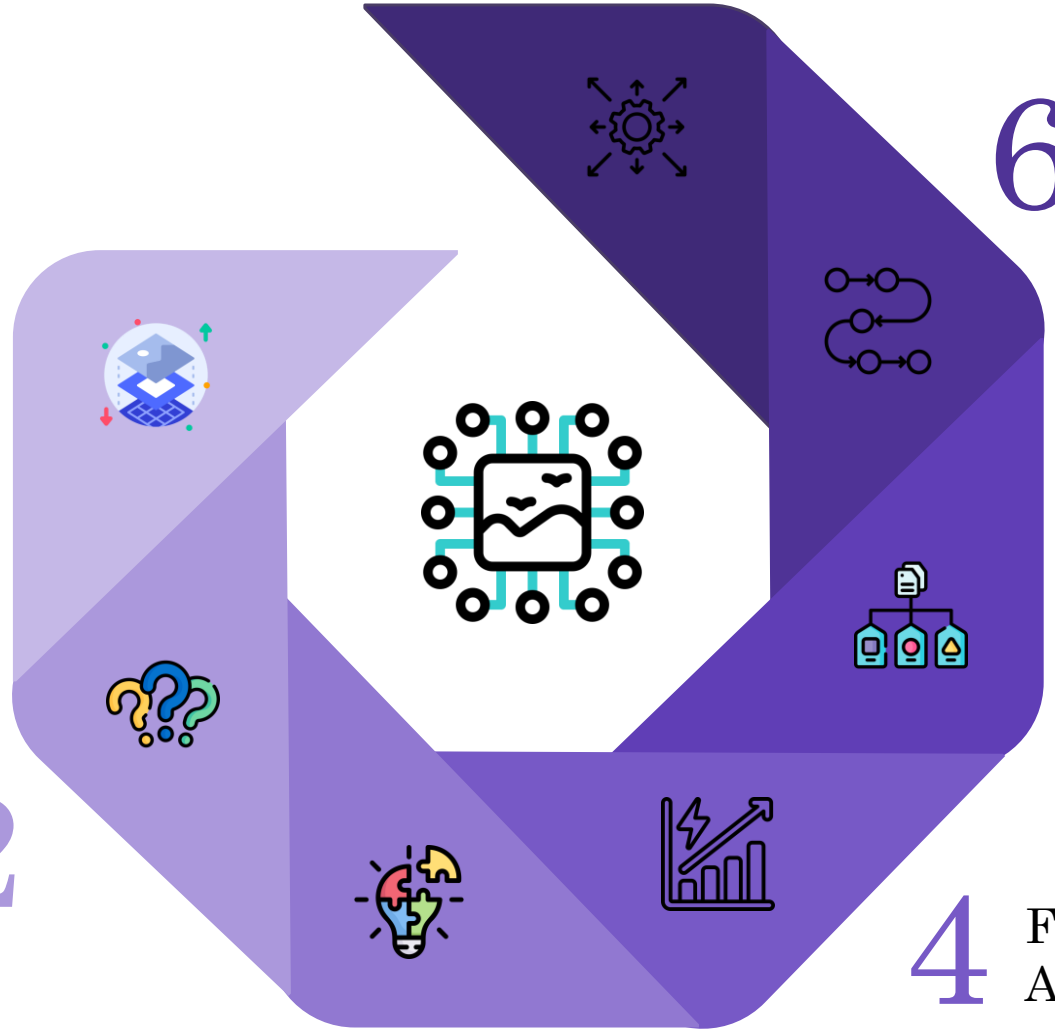
3 The Deep Learning Shift

4 Foundational Architectures (FCN)

5 Encoder-Decoder Models

6 Advanced Semantic Models

7 Instance & Real-Time Segmentation

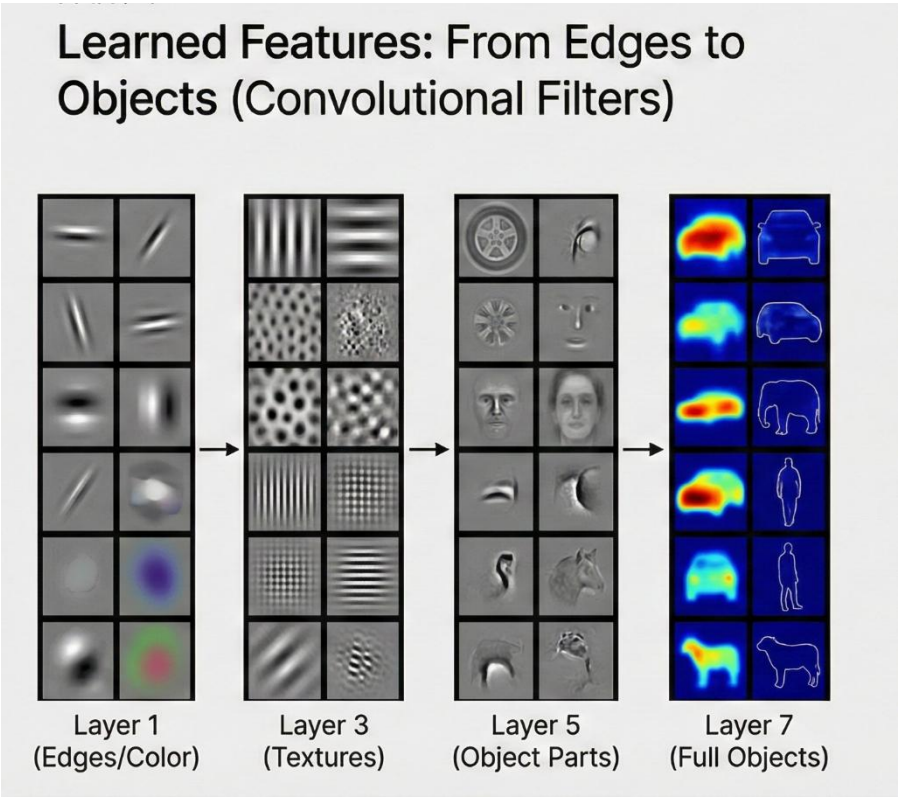


PART 4: DEEP LEARNING ERA

The Deep Learning Revolution - Why It Changed Everything

The Paradigm Shift

- **Classical Pipeline**
  - Hand-crafted features → Algorithm → Segmentation
  - (Humans design features, algorithm stays fixed)
- **Deep Learning Pipeline**
  - Raw pixels → Neural Network → Segmentation
  - (Network learns features + the segmentation rules)
- **Key Breakthroughs**



Learned Features

Networks learn hierarchical representations automatically:

- **Early layers → Edges, simple colors**
- **Mid layers → Textures, repeated patterns**
- **Deep layers → Object parts (eyes, wheels)**
- **Final layers → Full objects (faces, cars)**

❖ No manual feature engineering required.

PART 4: DEEP LEARNING ERA

The Deep Learning Revolution - Why It Changed Everything

End-to-End Learning

- One model trained with backpropagation
- Features + segmentation optimized together
- Removes hand-crafted rules

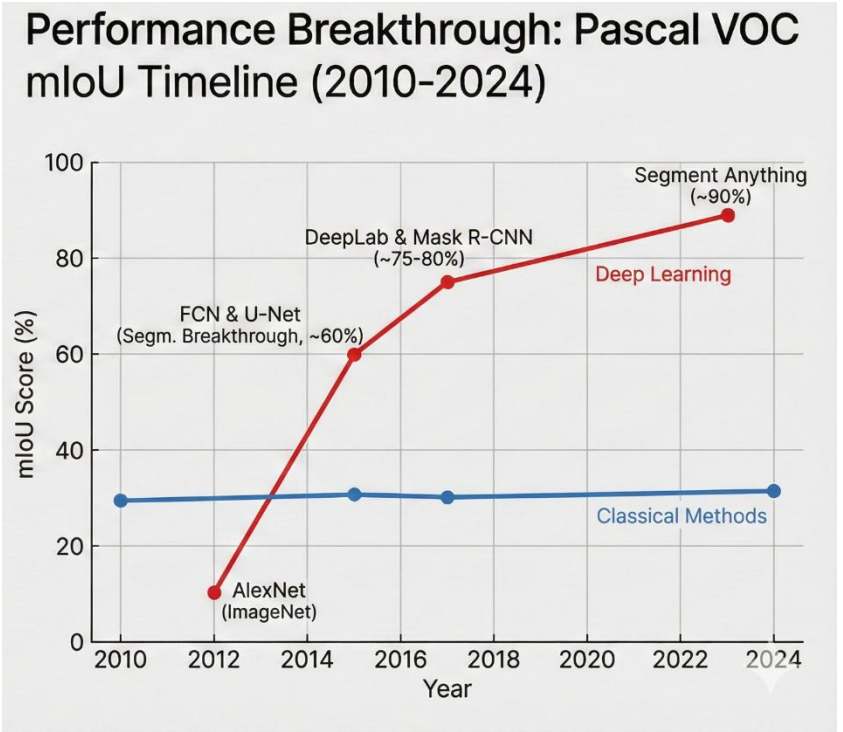
- **Scale With Data**
  - Performance improves with more data.
    Classical methods plateau early.

Transfer Learning

- Models pre-trained on massive datasets (ImageNet, COCO)
- Fine-tune on smaller tasks with strong results

What Enabled This?

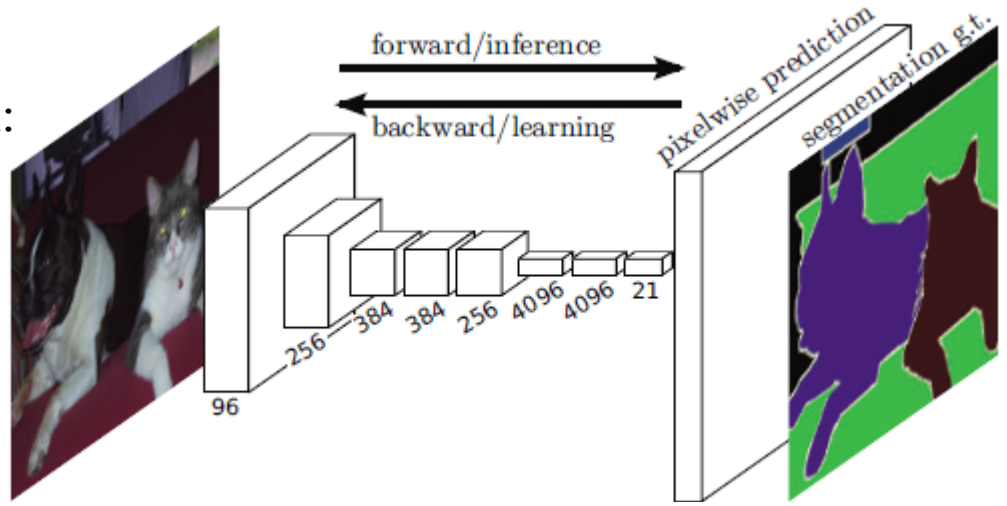
- **GPU computing** (massive parallel processing)
- **Large datasets** (ImageNet, COCO)
- **Better architectures** (CNNs, skip connections)
- **Improved training techniques** (batch norm, dropout, Adam)



Fully Convolutional Networks (FCN)

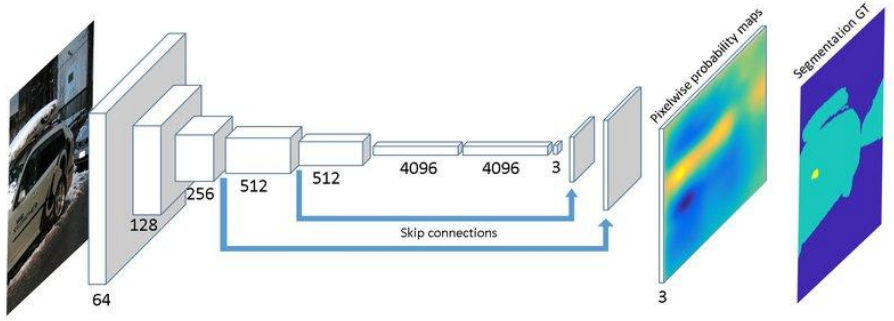
Why Regular CNNs Fail for Segmentation

- Classification networks compress the image too much:
  - Input → Convs → Pooling → **Flatten**
  - Fully connected layers destroy spatial layout
  - Output is a **single label**, not per-pixel predictions
- This makes them useless for dense tasks like segmentation.



The FCN Breakthrough (Long, Shelhamer, Darrell, 2015)

- **Remove Fully Connected Layers**
  - Replace them with **convolutions**, keeping spatial structure intact.
- **Predict a Heatmap**
  - Input: **H × W × 3**
  - After down sampling: **H/32 × W/32 × channels**
  - Apply 1×1 conv per location → **class scores per pixel**
  - Up sample back to H×W to produce the final mask

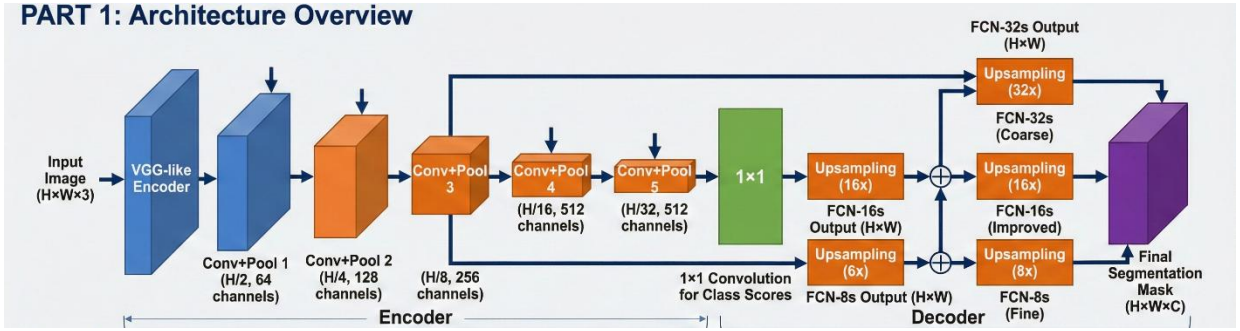


Fully Convolutional Networks (FCN)

Key Innovations

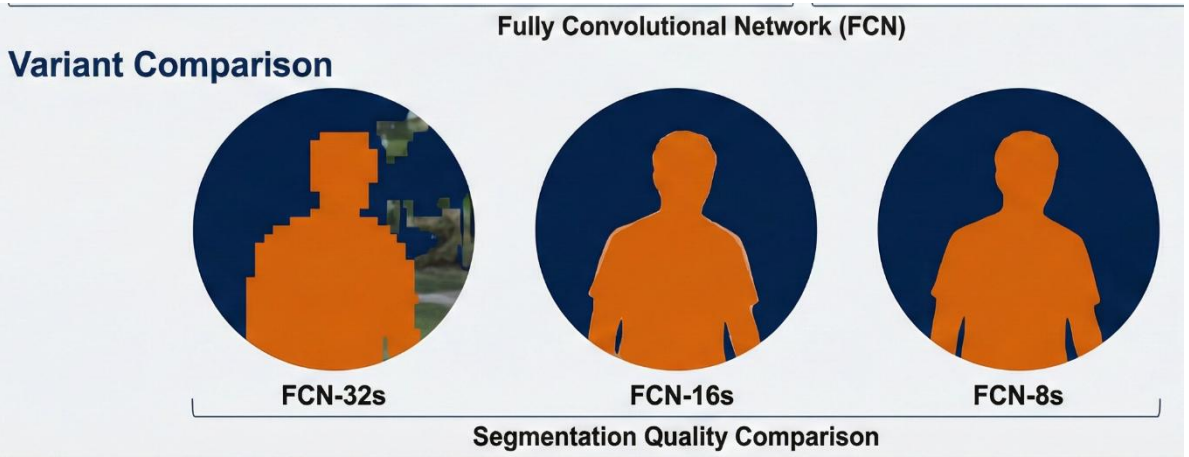
Fully Convolutional Architecture

- No dense layers
- Preserves spatial arrangement
- Works on any image size



Skip Connections

- Combine:
  - **Deep, semantic features** (coarse)
  - **Shallow, spatial features** (fine)
- This improves boundary quality.



- **Variants:**
  - **FCN-32s:** coarse result
  - **FCN-16s:** adds pool4 skip
  - **FCN-8s:** adds pool3 skip → **best boundaries**

Transfer Learning

- Initialize encoder with pretrained **VGG/ResNet** to stabilize training.

U-Net – The Medical Imaging Favorite

Why Medical Imaging Needed Something New

- Medical images often have:
  - Very **little training data** (hundreds, not millions)
  - **Life-critical boundaries** (tiny errors matter)
  - Mostly **grayscale scans** (CT, MRI, microscopy)
  - Strong need for **pixel-accurate** segmentation
- U-Net was designed specifically to solve these constraints.

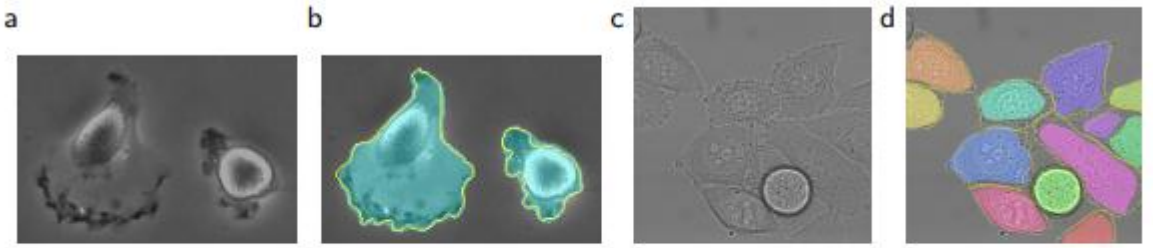
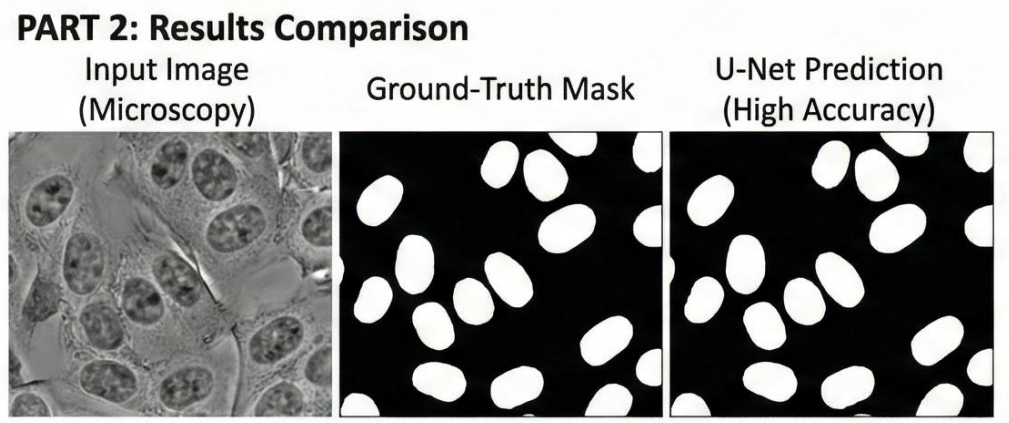
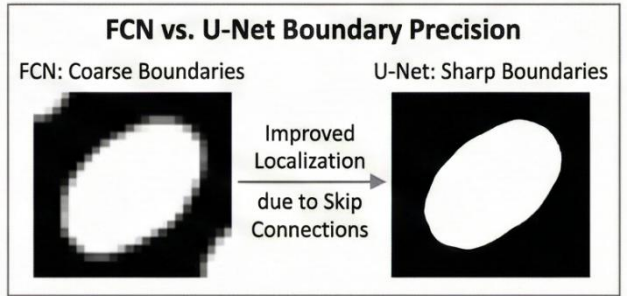


Fig. 4. Result on the ISBI cell tracking challenge. (**a**) part of an input image of the "PhC-U373" data set. (**b**) Segmentation result (cyan mask) with manual ground truth (yellow border) (**c**) input image of the "DIC-HeLa" data set. (**d**) Segmentation result (random colored masks) with manual ground truth (yellow border).





U-Net – The Medical Imaging Favorite

- **U-Net Architecture (Ronneberger et al., 2015)**

Encoder – Contracting Path

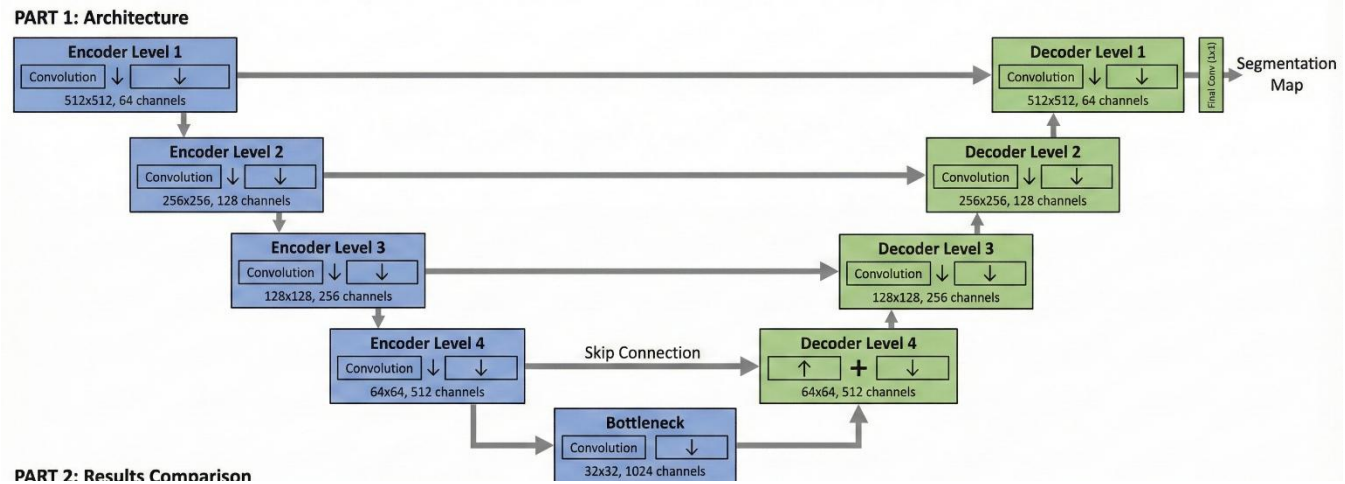
- Repeated blocks of:
  - Convolution → Convolution → MaxPool
  - Extracts context
  - Shrinks spatial resolution

Decoder – Expansive Path

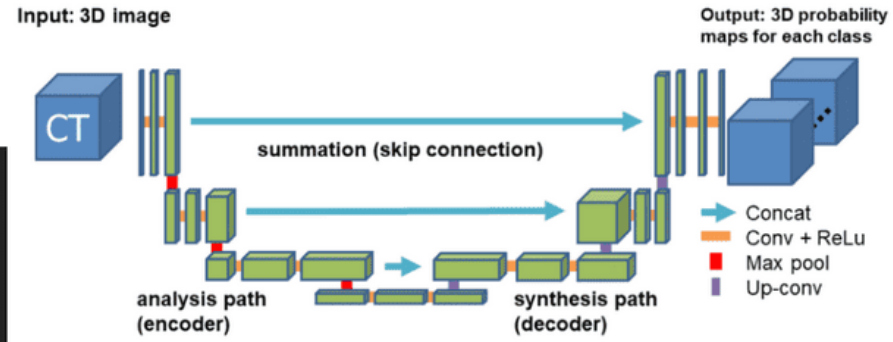
- Repeated blocks of:
  - Up-convolution
  - **Concatenation with encoder feature maps**
  - Convolution → Convolution
  - Recovers spatial precision

Bottleneck

- Deepest layer combining:
  - High-level semantic representation
  - Smallest spatial resolution




```
def forward(self, x):
x1 = self.inc(x)
x2 = self.down1(x1)
x3 = self.down2(x2)
x4 = self.down3(x3)
x5 = self.down4(x4)
x = self.up1(x5, x4)
x = self.up2(x, x3)
x = self.up3(x, x2)
x = self.up4(x, x1)
x = self.outc(x)
return x
```



U-Net – The Medical Imaging Favorite

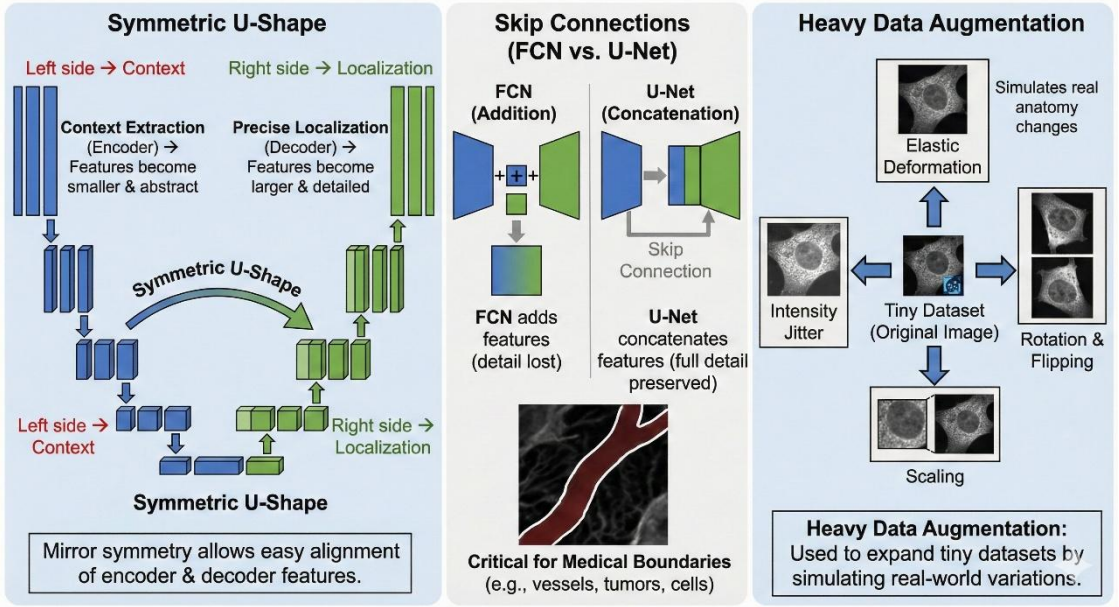
Key Features

Symmetric U-Shape

- Left side → **Context extraction**
- Right side → **Precise localization**
- The mirror symmetry makes it easy to align encoder and decoder features.

Skip Connections (Better Than FCN's)

- FCN adds features
- **U-Net concatenates features**
- This preserves **full low-level detail**
- Critical for medical boundaries like vessels, tumors, cells



Heavy Data Augmentation

- Used because datasets are tiny:
  - Elastic deformation (simulates real anatomy changes)
  - Rotation and flipping
  - Scaling
  - Intensity jitter

Weighted Loss for Hard Regions

Gives extra weight to:

- Object borders
- Thin structures
- Touching cells

Helps prevent merging or missing fine details.

U-Net – The Medical Imaging Favorite

Why U-Net Became the Standard

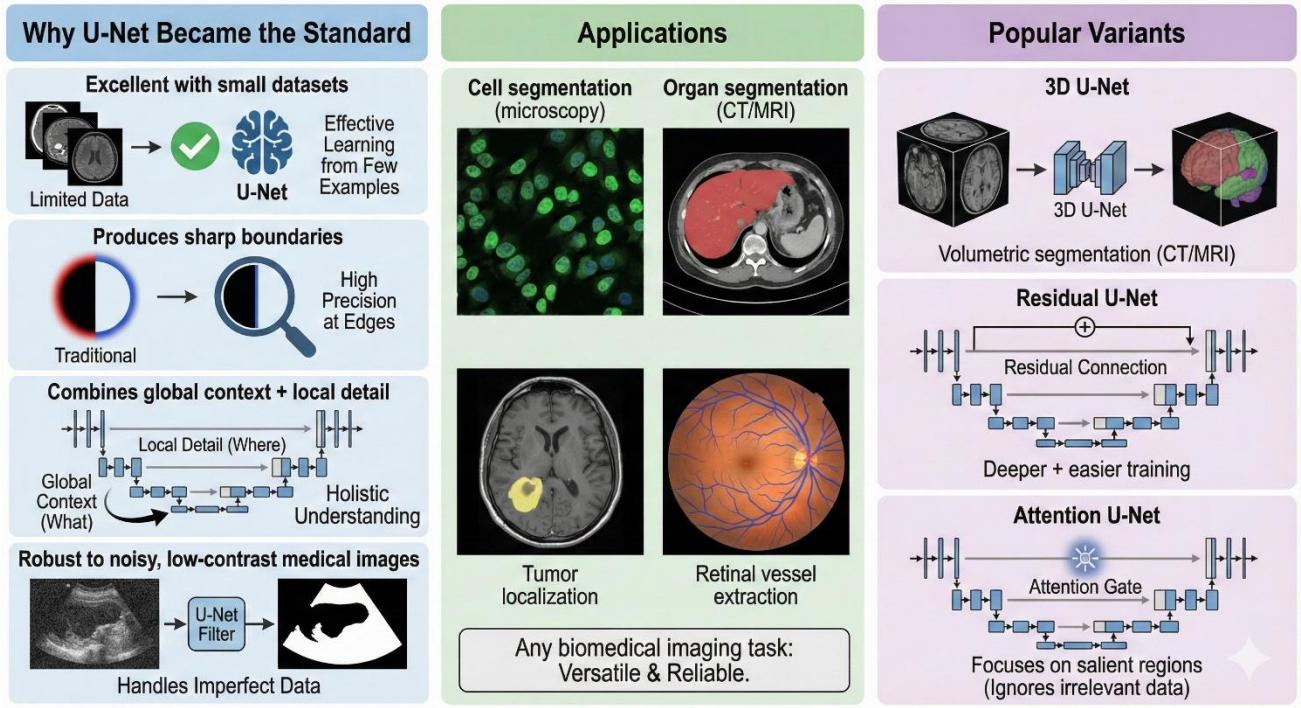
- Excellent with **small datasets**
- Produces **sharp boundaries**
- Combines **global context + local detail**
- Robust to noisy, low-contrast medical images

Applications

- Cell segmentation (microscopy)
- Organ segmentation (CT/MRI)
- Tumor localization
- Retinal vessel extraction
- Any biomedical imaging task

Popular Variants

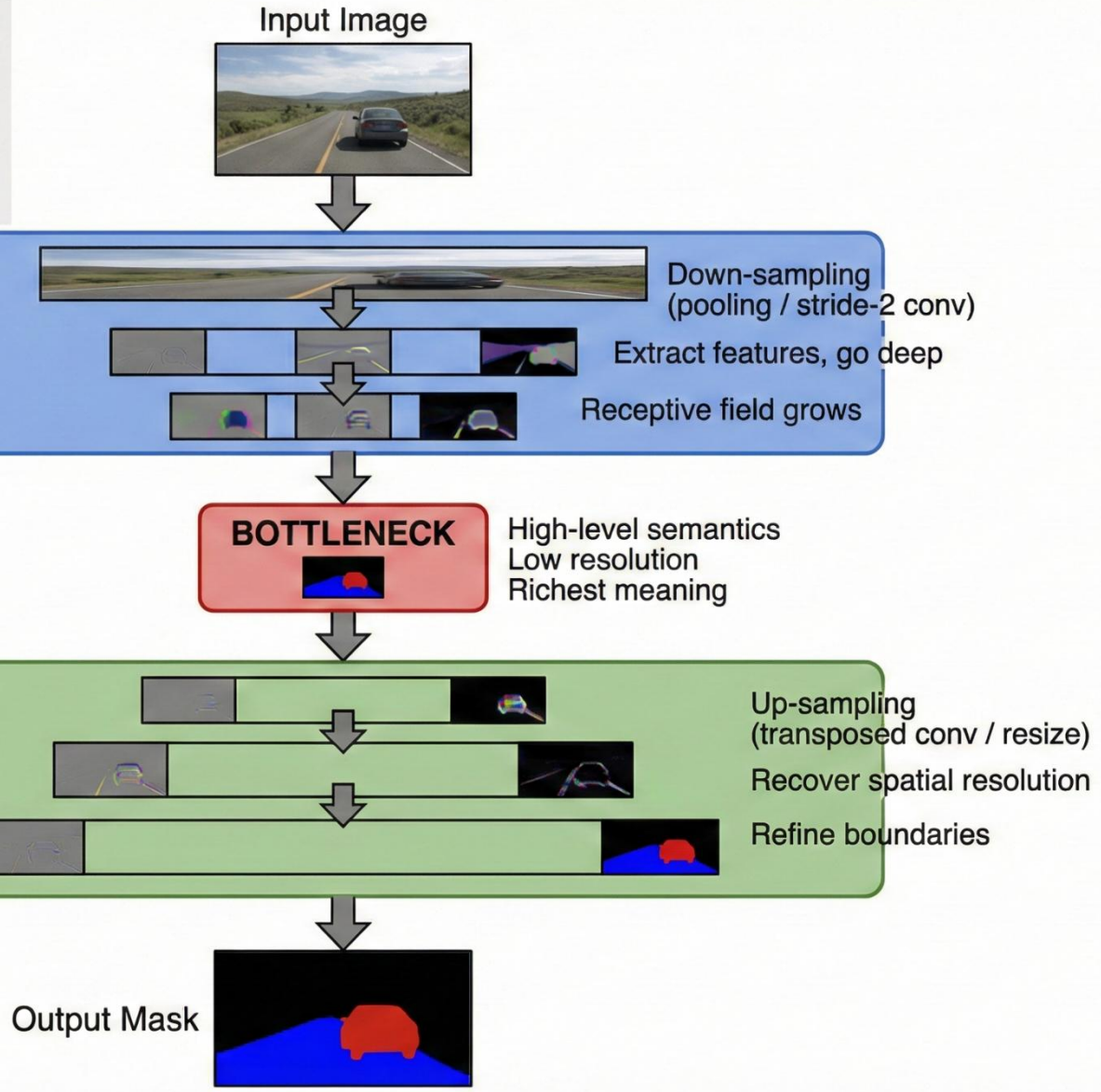
- **3D U-Net** – volumetric segmentation for CT/MRI
- **Residual U-Net** – deeper + easier training
- **Attention U-Net** – focuses on salient regions



Encoder–Decoder Architectures

Why This Pattern Exists: All strong segmentation models follow the same backbone idea.

- FCN
- U-Net
- SegNet
- DeepLab
- PSPNet



Encoder–Decoder Architectures

Key Design Choices (What You Trade Off)

Encoder

- **VGG / ResNet / EfficientNet / MobileNet**
- **Trade-off:** Depth vs speed vs memory

Decoder

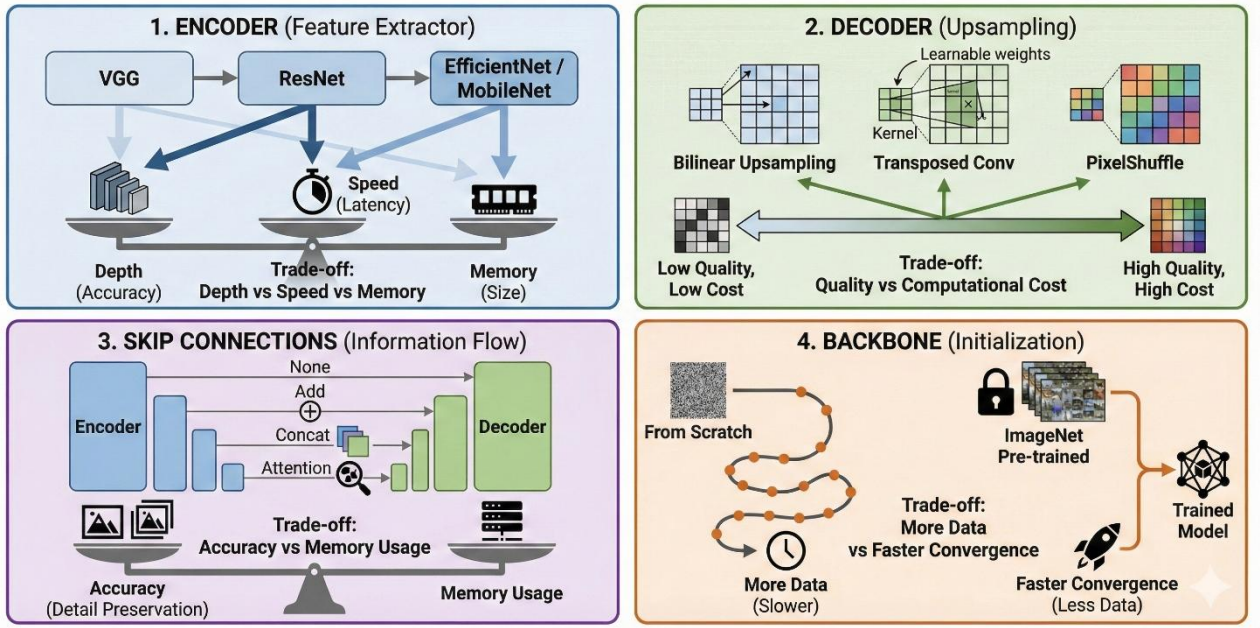
- **Bilinear up-sampling / Transposed Conv / PixelShuffle**
- **Trade-off:** Quality vs computational cost

Skip Connections

- **None / Add / Concat / Attention**
- **Trade-off:** Accuracy vs memory usage

Backbone

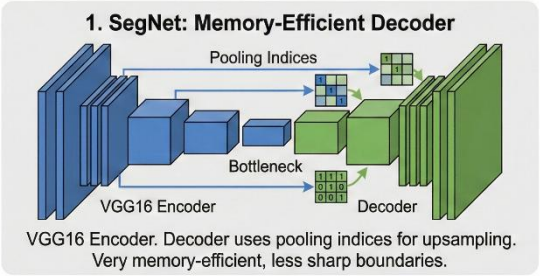
- **From scratch / ImageNet pre-trained**
- **Trade-off:** More data vs faster convergence

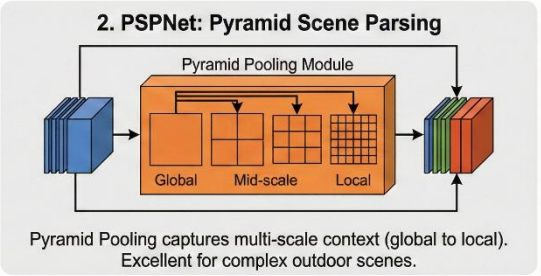


Encoder–Decoder Architectures

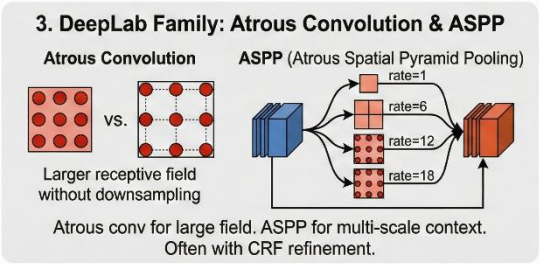
Representative Architectures

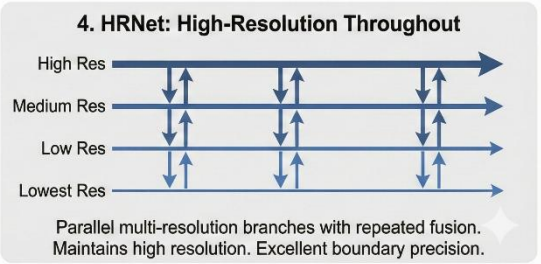
- **SegNet**
  - Encoder = VGG16
  - Decoder uses **pooling indices** instead of learnable upsampling
  - Very memory-efficient, but not as sharp as U-Net





- **PSPNet (Pyramid Scene Parsing)**
  - Introduced **Pyramid Pooling Module**
  - Captures:
    - Global context
    - Mid-scale context
    - Local detail
  - Works extremely well for complex outdoor scenes (Cityscapes).





- **DeepLab Family**
  - **Atrous (Dilated) convolutions** → large receptive field without down-sampling
  - **ASPP (Atrous Spatial Pyramid Pooling)** → multi-scale context
  - Often used with CRF for refinement (older versions)

HRNet

- Holds **high resolution throughout** the network
- Parallel branches at:
  - High res
  - Medium res
  - Low res
- Fuses them repeatedly
- Excellent boundary precision

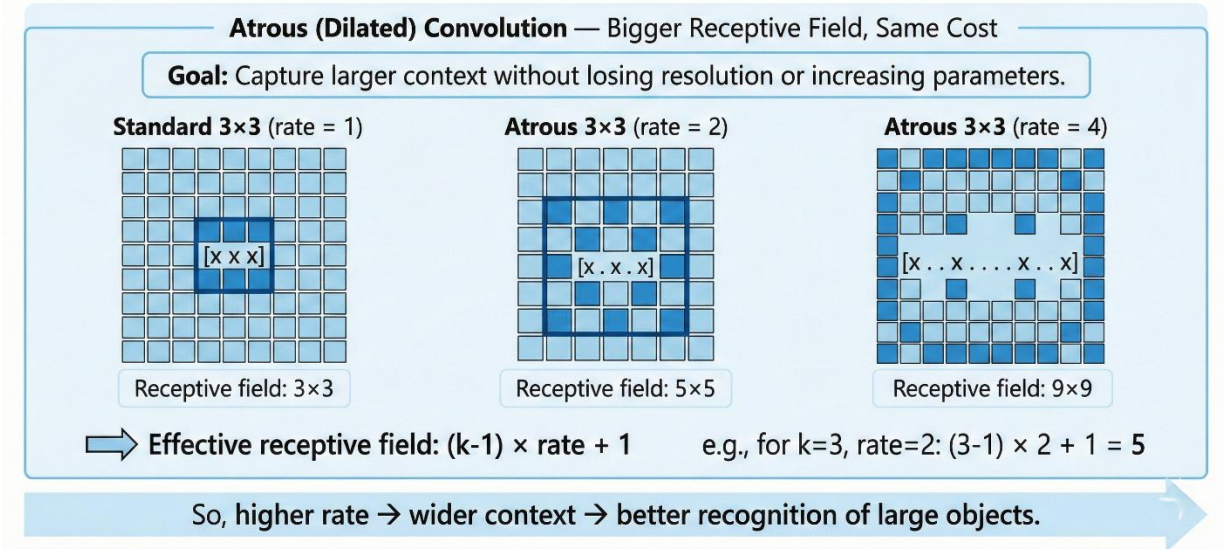
DeepLabv3+ — State-of-the-Art Semantic Segmentation

Why It Matters

- DeepLabv3+ combines dilated convolutions, ASPP, and a clean decoder, giving excellent accuracy on complex datasets (Cityscapes, Pascal VOC).

Atrous (Dilated) Convolution — Bigger Receptive Field, Same Cost

- Goal: capture larger context without losing resolution or increasing parameters.
- Standard 3×3 (rate = 1): [x x x]
- Atrous 3×3 (rate = 2): [x . x . x]
- Atrous 3×3 (rate = 4): [x . . . x . . . x]
- Effective receptive field:
  $(k-1) \times rate + 1 (k-1) \times rate + 1$
- So, higher rate → wider context → better recognition of large objects.



DeepLabv3+ — State-of-the-Art Semantic Segmentation

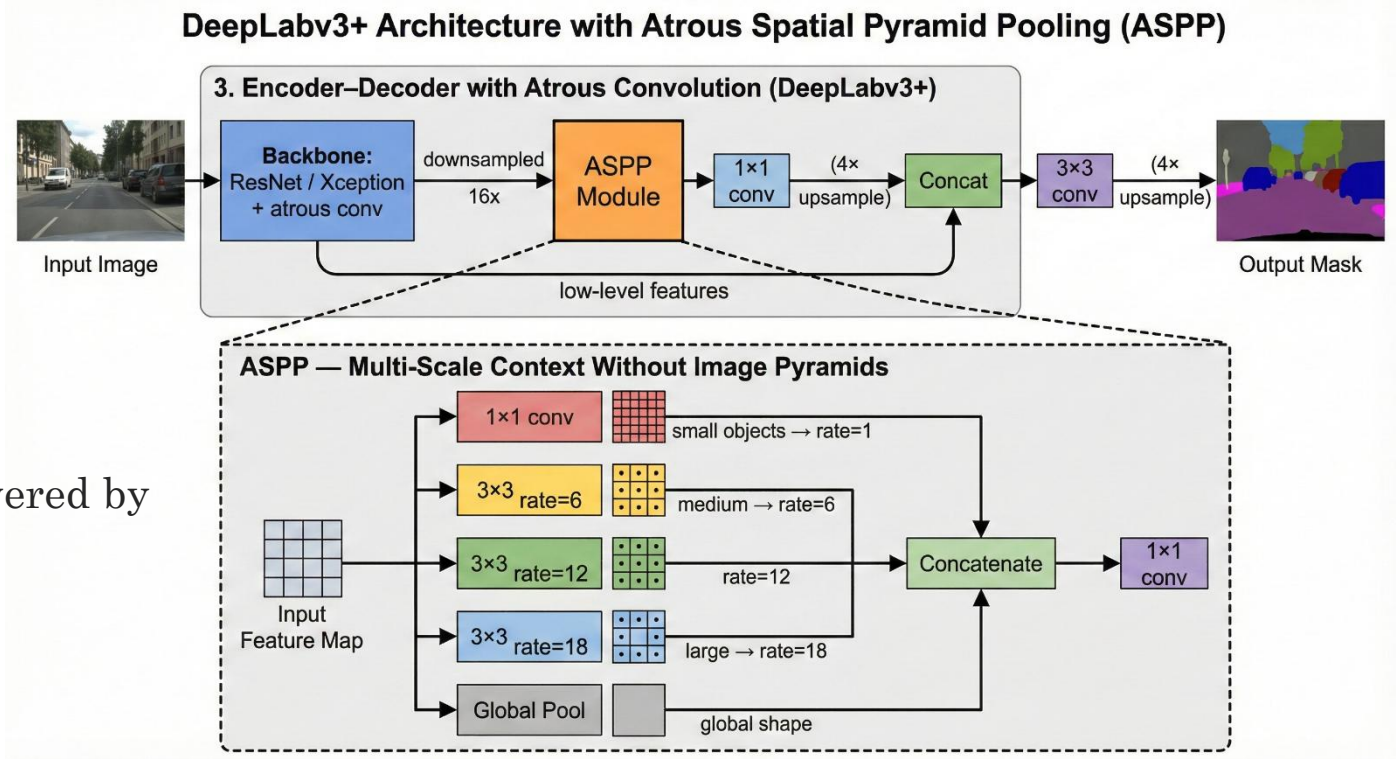
ASPP — Multi-Scale Context Without Image Pyramids

- ASPP does **simultaneous multi-scale analysis**:
  - small objects → rate=1
  - medium → rate=6, 12
  - large → rate=18
  - global shape → global pooling

- **Encoder–Decoder with Atrous Convolution (DeepLabv3+)**
  - A refined U-Net-like structure but powered by atrous convs.

Why It Works

- Multi-scale understanding (ASPP)
- Huge receptive field without blurring
- Decoder restores sharp edges
- Strong performance on fine boundaries (roads, poles, pedestrians)



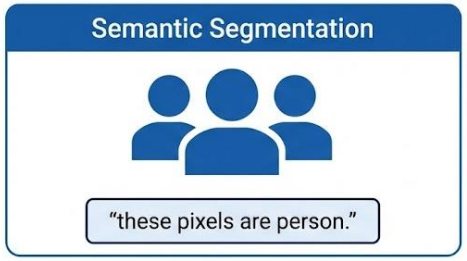
Instance Segmentation — Distinguishing Individuals

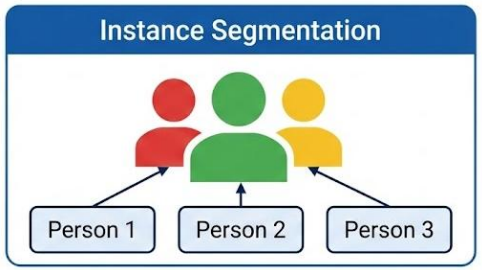
What Problem Does It Solve?

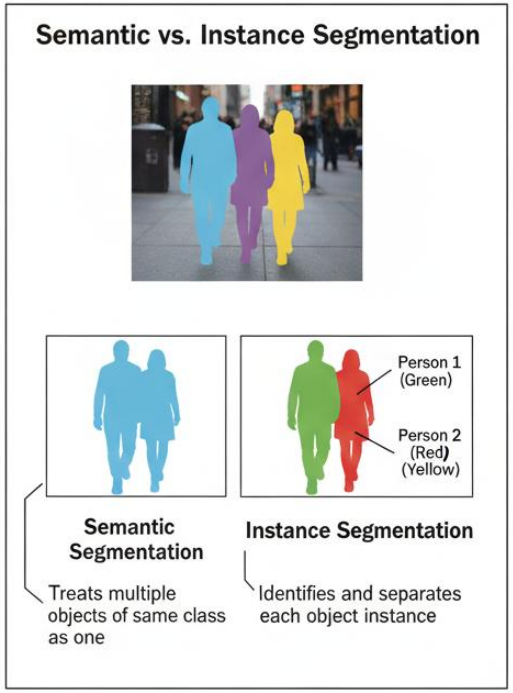
- Semantic segmentation only says **"these pixels are person."**
  Instance segmentation must say:
  - **Person 1**
  - **Person 2**
  - **Person 3**
- Even when they overlap or vary in size.

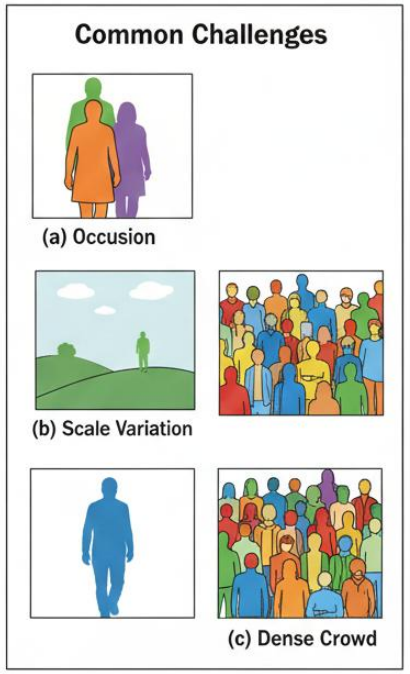
Challenges

- **Changing instance count** (1 → 100+)
- **Occlusions** (one object covering another)
- **Scale variation** (tiny vs huge objects)
- **Crowded scenes**





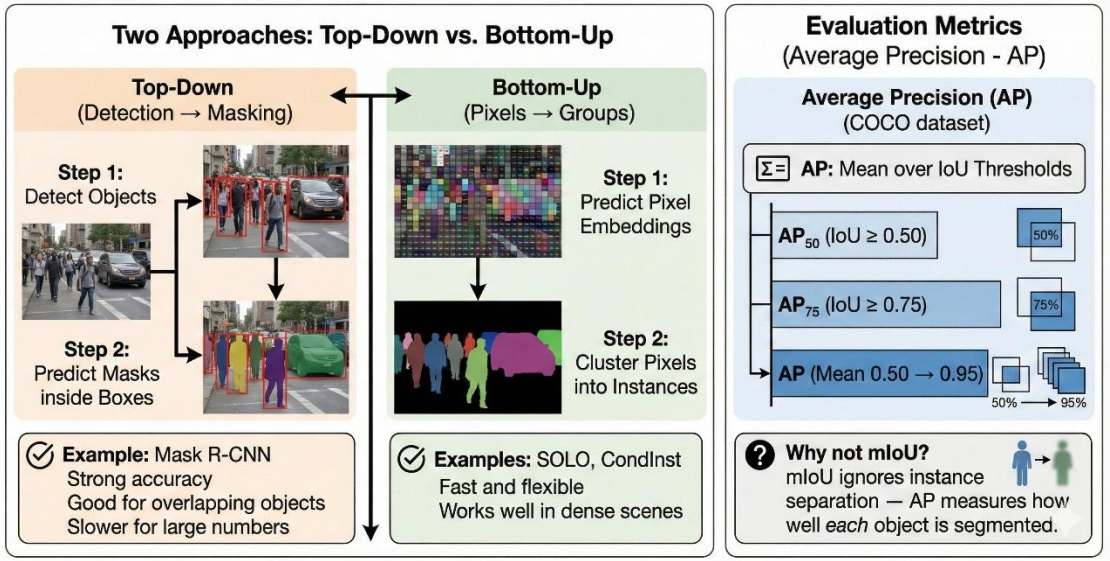




Instance Segmentation — Distinguishing Individuals

Two Approaches

- **Top-Down (Detection → Masking)**
  - **Steps:**
    - Detect objects (bounding boxes)
    - Predict a precise mask inside each box
  - **Example: Mask R-CNN**
    - Strong accuracy
    - Good for overlapping objects
    - Slower for large numbers of instances
- **Bottom-Up (Pixels → Groups)**
  - **Steps:**
    - Predict pixel embeddings/features
    - Cluster pixels into separate instances
  - **Examples:** SOLO, CondInst
    - Fast and flexible
    - Works well in dense scenes



Evaluation Metrics

Average Precision (AP)

Used in COCO dataset:

- **AP_{50}:** IoU ≥ 0.50
- **AP_{75}:** IoU ≥ 0.75
- **AP:** Mean over IoU 0.50 → 0.95

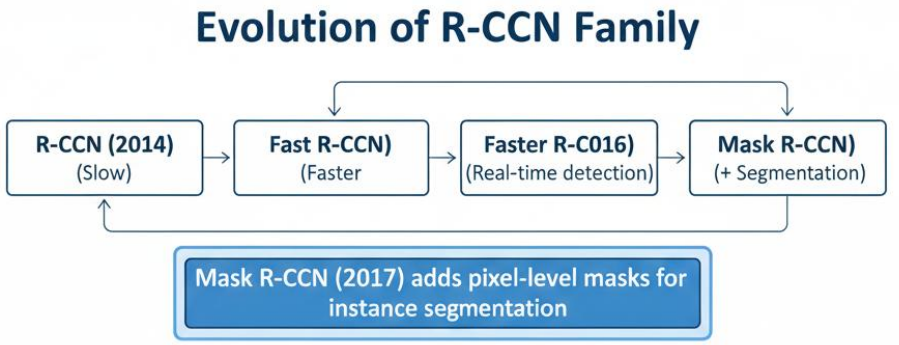
Why not mIoU?

mIoU ignores instance separation — AP measures how well **each object** is segmented.

Mask R-CNN: The Standard for Instance Segmentation

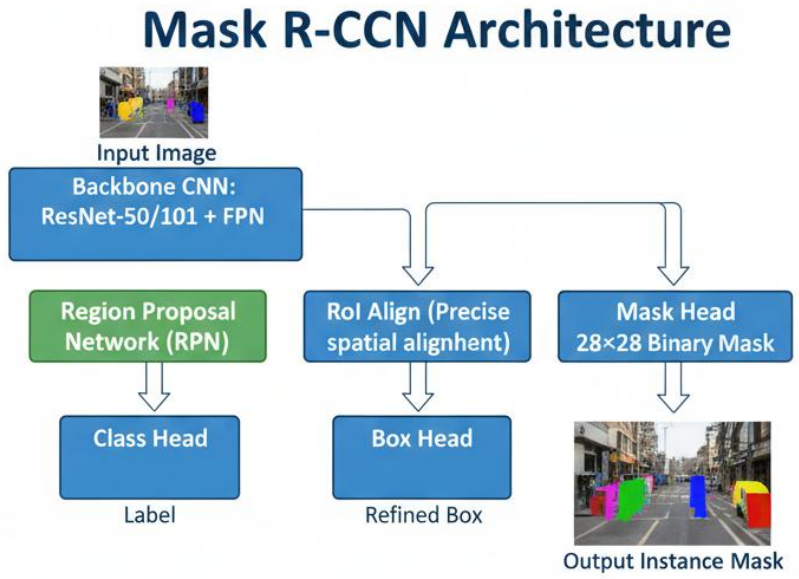
Evolution (How We Got Here)

- **R-CNN (2014):** Accurate but slow
- **Fast R-CNN (2015):** Faster training
- **Faster R-CNN (2016):** Real-time proposals (RPN)
- **Mask R-CNN (2017):** Adds **pixel-level masks** → instance segmentation champion



Architecture Overview

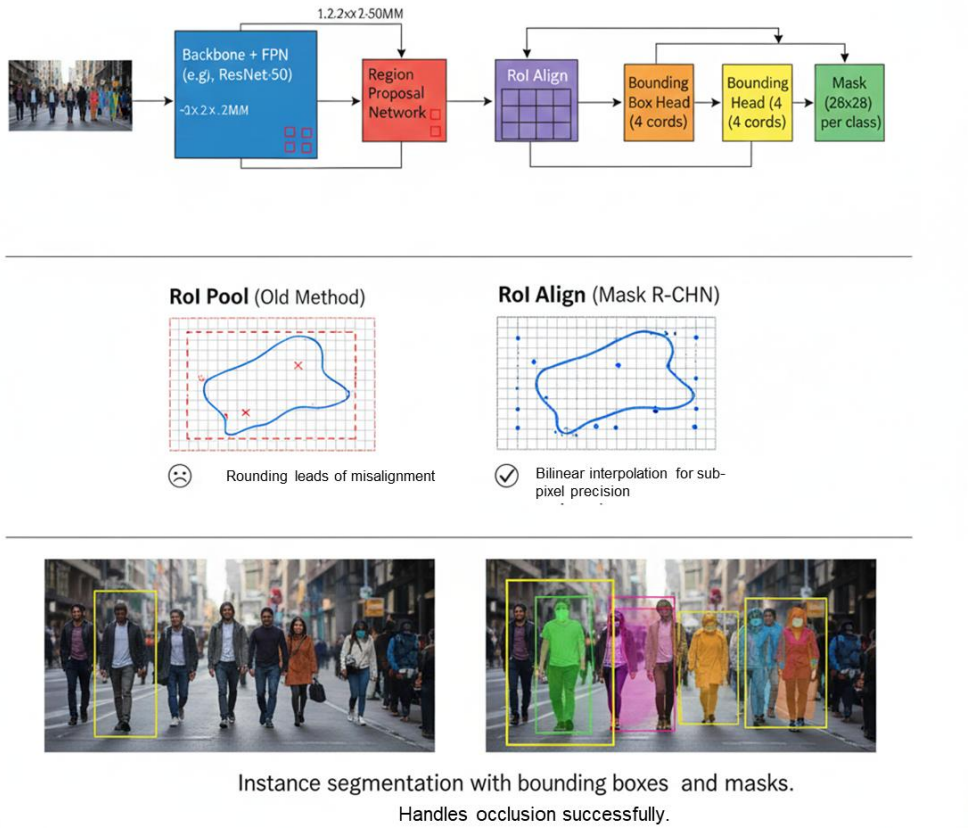
- **Input Image**
  → Backbone (ResNet-50/101 + FPN)
  → Region Proposal Network (RPN)
  → RoIs
  → RoI Align
  → Three Parallel Heads:
    - **Class Head:** Predicts object category
    - **Box Head:** Refines bounding box
    - **Mask Head:** Predicts **28×28 binary mask** per instance



Mask R-CNN: The Standard for Instance Segmentation

Key Components

- **Feature Pyramid Network (FPN)**
  - Combines high-resolution + high-semantic features
  - Gives strong multi-scale detection
- **Region Proposal Network (RPN)**
  - Generates ~1,000 candidate object regions
  - Fast and fully convolutional
- **RoI Align (Major Upgrade)**
  - **Problem (RoI Pool):**
    - Rounds coordinates → misalignment → bad masks
  - **Fix (RoI Align):**
    - **No rounding**
    - Uses **bilinear interpolation**
    - Results in **sub-pixel accuracy**
    - Essential for high-quality masks



Mask Branch

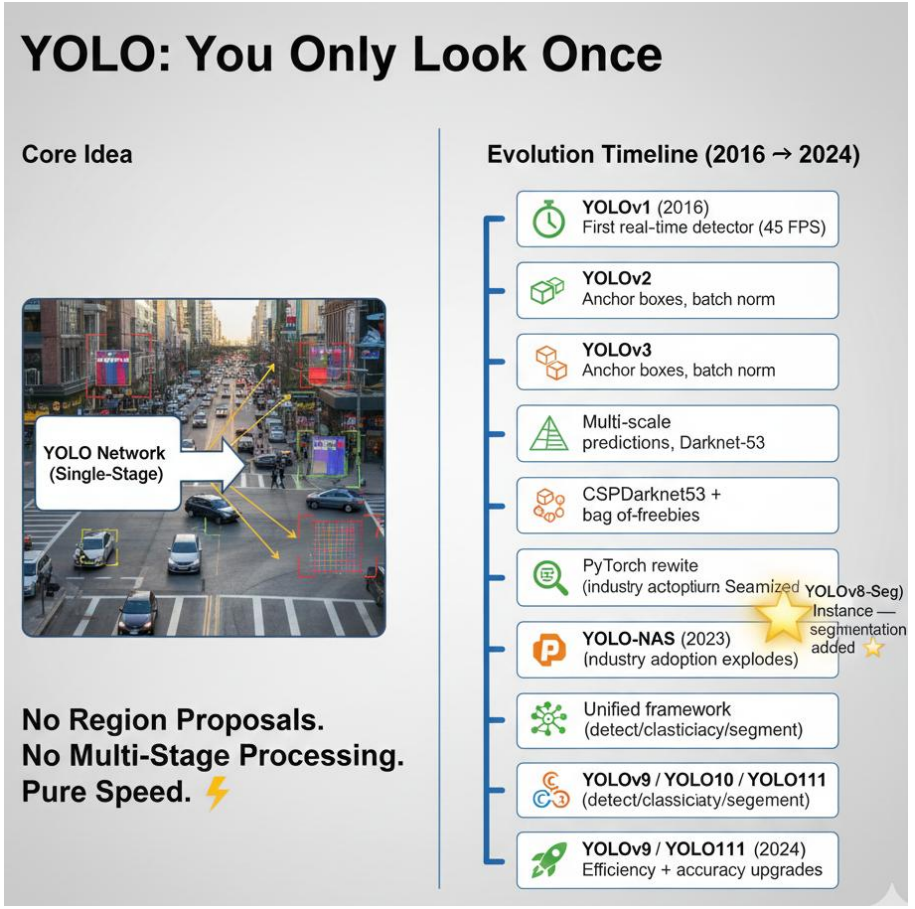
- Runs **in parallel** to class + box branches
- FCN-based small decoder
- Outputs **28×28 mask** for each class
- Refined to full resolution at the end

YOLO Evolution: From Detection to Segmentation

Core Idea

- **YOLO = You Only Look Once**
  A **single-stage** network that predicts bounding boxes, classes, and (now) masks **in one forward pass**.
  No region proposals. No multi-stage processing. Pure speed.

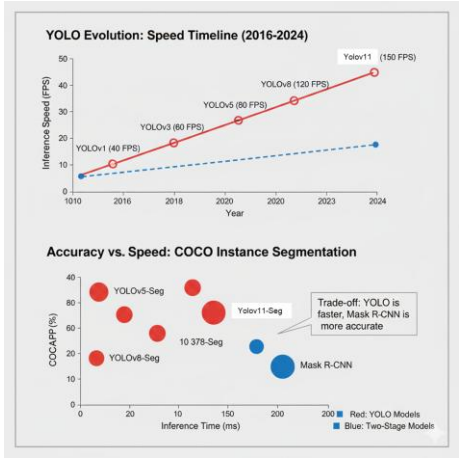
- **Evolution Timeline (2016 → 2024)**
  - **YOLOv1 (2016)** — First real-time detector (45 FPS)
  - **YOLOv2 (2017)** — Anchor boxes, batch norm
  - **YOLOv3 (2018)** — Multi-scale predictions, Darknet-53
  - **YOLOv4 (2020)** — CSPDarknet53 + bag-of-freebies
  - **YOLOv5 (2020)** — PyTorch rewrite (industry adoption explodes)
  - **YOLO-NAS (2023)** — Neural Architecture Search optimized
  - **YOLOv8 (2023)** — Unified framework for detect/classify/segment
  - **YOLOv8-Seg (2023)** — **Instance segmentation** added
  - **YOLOv9 / YOLOv10 / YOLOv11 (2024)** — Efficiency + accuracy upgrades



YOLO Evolution: From Detection to Segmentation

Why YOLO for Segmentation?

- **Fast:** 30–60 FPS on a mid-range GPU
- **Lightweight:** Runs on laptops, embedded devices, even mobile
- **Simple:** One network, one forward pass
- **Practical:** Ideal for real-time tasks (CCTV, robotics, drones, AR)



YOLO vs. Mask R-CNN (Direct Comparison)

Feature	**Mask R-CNN**	**YOLOv8-Seg**
Architecture	Two-stage	One-stage
Speed (FPS)	5–7	30–60
Accuracy (COCO AP)	~37.1%	~36%
Complexity	High	Medium
Deployment	Server	Edge/Mobile

Trade-off:

- YOLO is **much faster**, slightly less accurate.
  Perfect for **real-time** systems where latency matters more than 1–2 AP.

YOLOv8 Segmentation: Architecture Overview

Pipeline

Input (640×640×3)

→**Backbone: CSPDarknet**

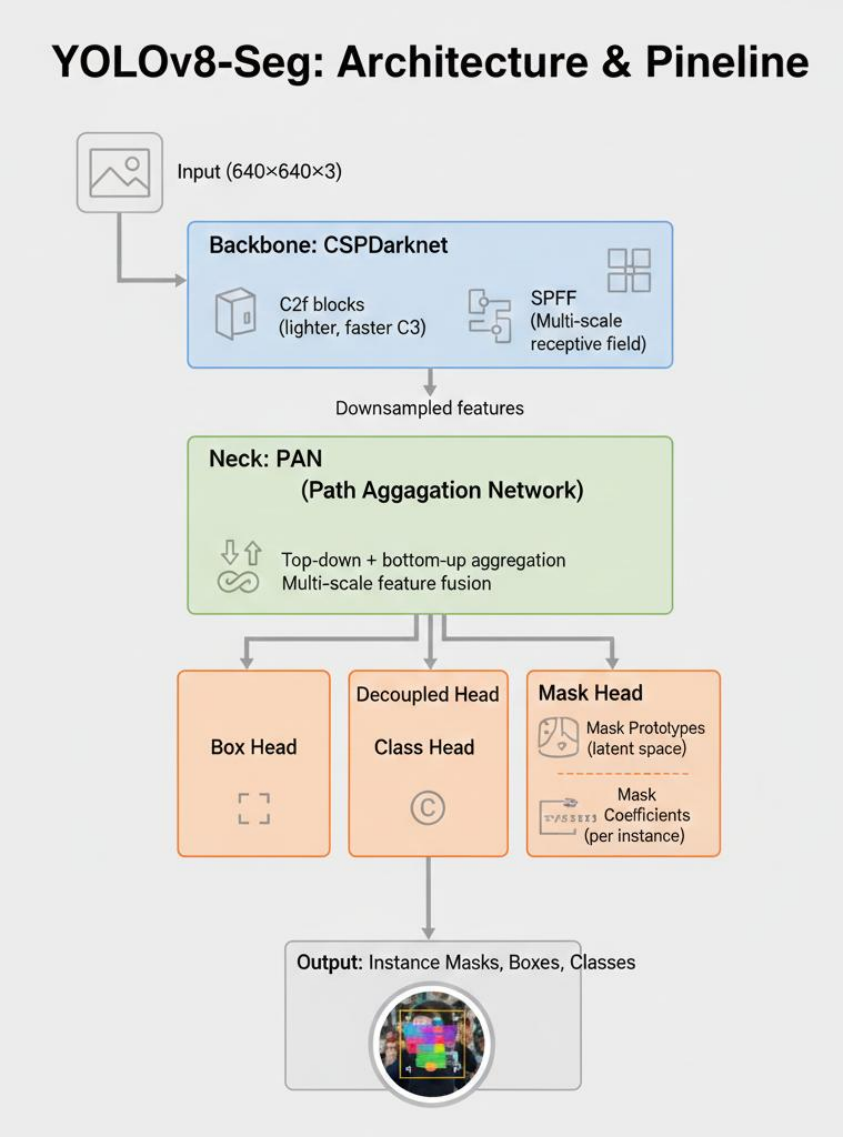
- C2f blocks (lighter, faster C3)
- SPPF for multi-scale receptive field

→**Neck: PAN**

- Top-down + bottom-up aggregation
- Multi-scale feature fusion

→**Decoupled Heads**

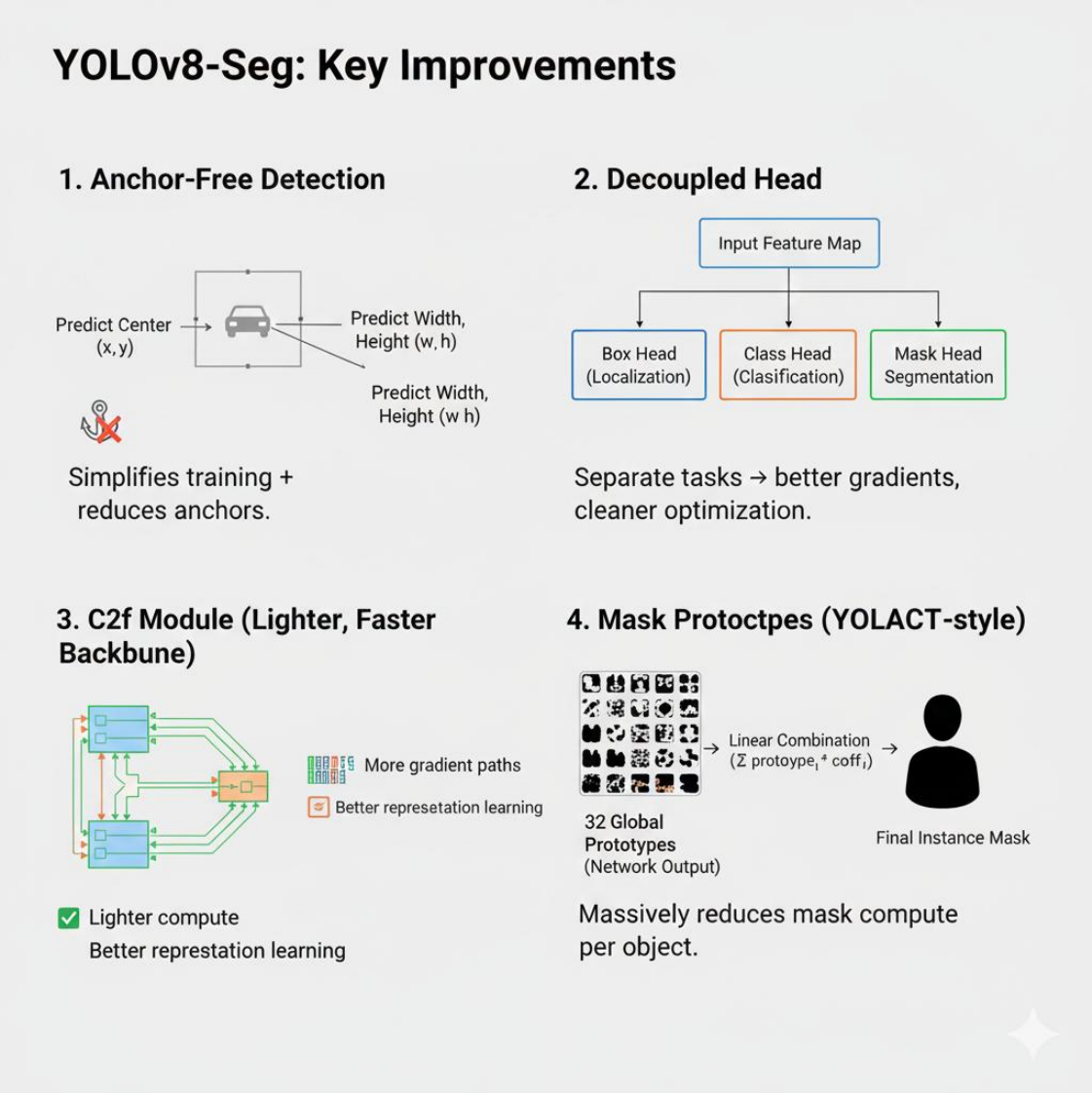
- **Box Head**
- **Class Head**
- **Mask Head** (prototype + coefficients)



YOLOv8 Segmentation: Architecture Overview

Key Improvements

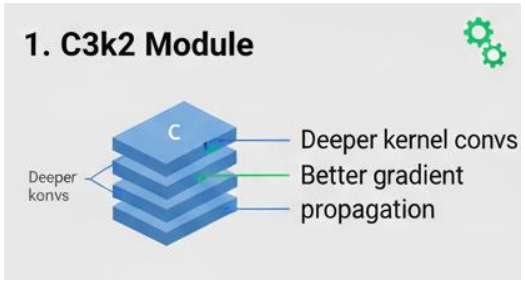
- **Anchor-Free Detection**
  - Predict center + width/height directly. Simplifies training + reduces anchors.
- **Decoupled Head**
  - Separate classification and localization → better gradients, cleaner optimization.
- **C2f Module**
  - Hybrid of CSPNet + ELAN:
    - more gradient paths
    - lighter compute
    - better representation learning
- **Mask Prototypes (YOLACT-style)**
  - Network predicts **32 global prototypes**
  - Each instance predicts **32 coefficients**
  - Linear combination → final mask
    Massively reduces mask compute per object.

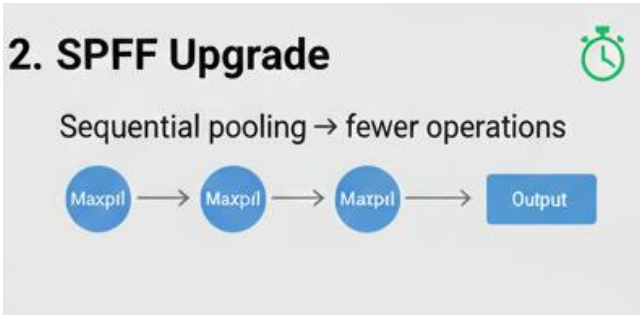


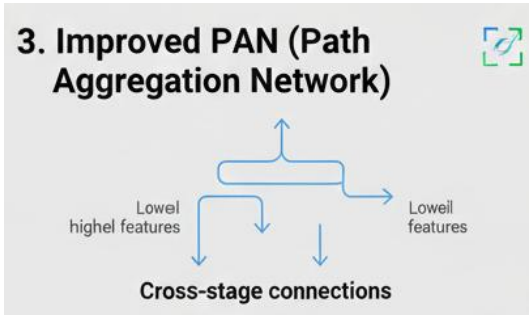
YOLOv11 Segmentation: Latest Improvements (2024)

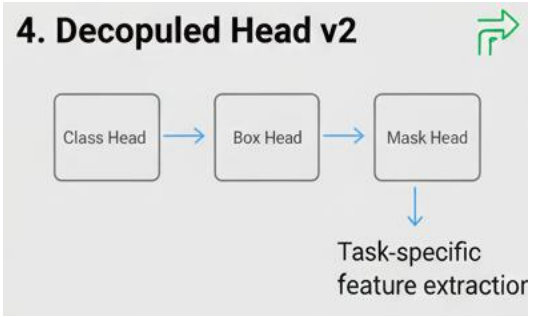
What's New in YOLOv11-Seg

- **C3k2 Module**
  - Enhanced version of C2f
  - Deeper kernel convs → stronger multi-scale representation
  - Better gradient propagation
  - More stable training on high-resolution images
- **SPPF Upgrade**
  - Same receptive field as YOLOv8
  - Sequential pooling → fewer operations
  - Lower latency, improved throughput
- **Improved PAN**
  - Stronger feature fusion
  - Added cross-stage connections
  - Dynamic feature selection for small objects









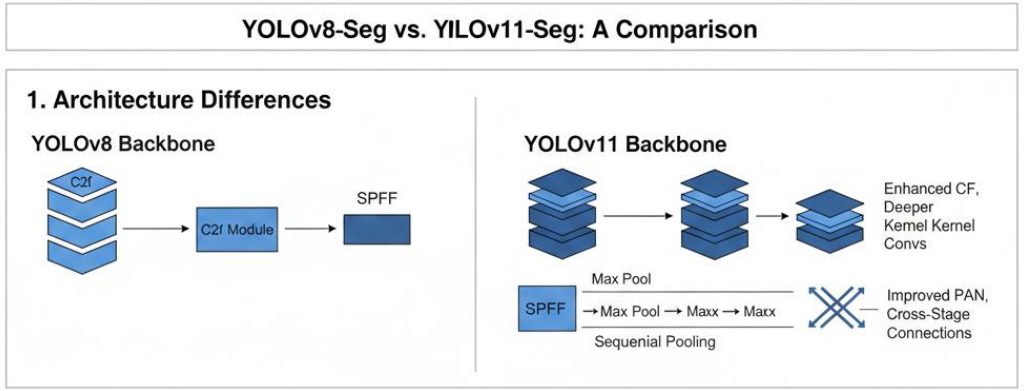
Decoupled Head v2

- Less parameter sharing between heads
- Task-specific feature extraction
- Cleaner gradients for box / class / mask

YOLOv11 Segmentation: Latest Improvements (2024)

Training Improvements

- **Better Augmentation**
  - Copy-paste
  - Large-scale jitter
  - Auto-augment policies
- **Upgraded Loss Functions**
  - Varifocal loss → classification
  - Alpha-IoU → box regression
  - Dice loss → mask prediction



Performance Gains

Metric	YOLOv8-Seg	YOLOv11-Seg	Δ
AP (box)	52.3%	53.4%	+1.1%
AP (mask)	42.6%	43.9%	+1.3%
Speed	35 FPS	40 FPS	+14%
Params	27.3M	25.9M	−5%

Panoptic Segmentation – Full Scene Understanding

What It Does:

- Panoptic segmentation combines **semantic segmentation** (labeling regions) and **instance segmentation** (separating objects).
  Every pixel gets both **a class** and **an instance ID** (if needed).

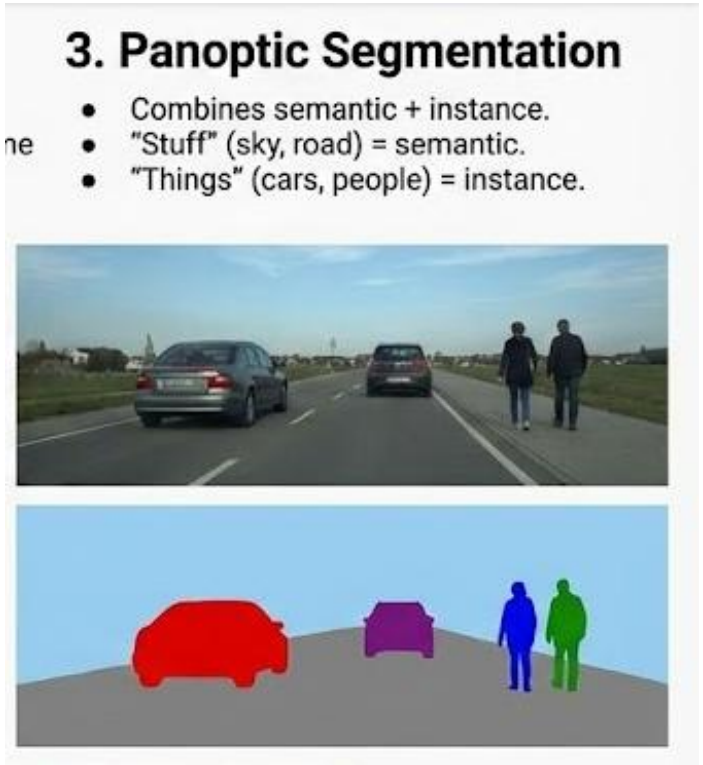
Two Types of Classes

❖**Stuff (No instances)**

- Large continuous regions:
  - sky
  - road
  - grass
  - water
- Handled by: **Semantic segmentation**

❖**Things (Countable objects)**

- Objects you can separate:
  - person
  - car
  - cat
  - bicycle
- Handled by: **Instance segmentation**



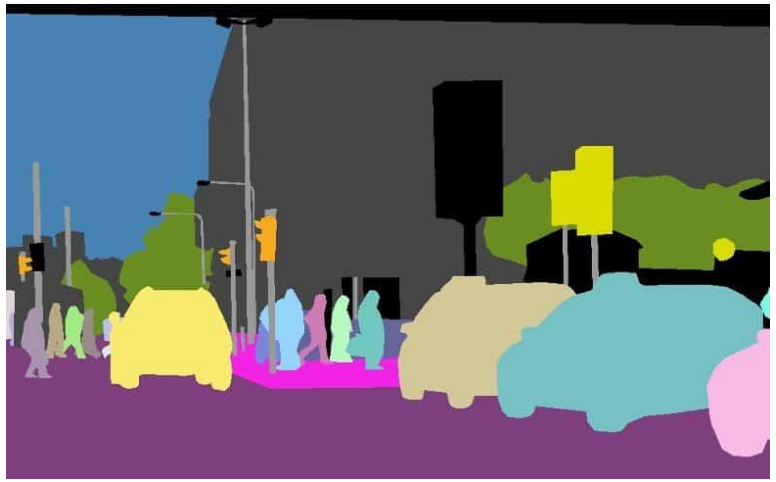
Panoptic Segmentation – Full Scene Understanding

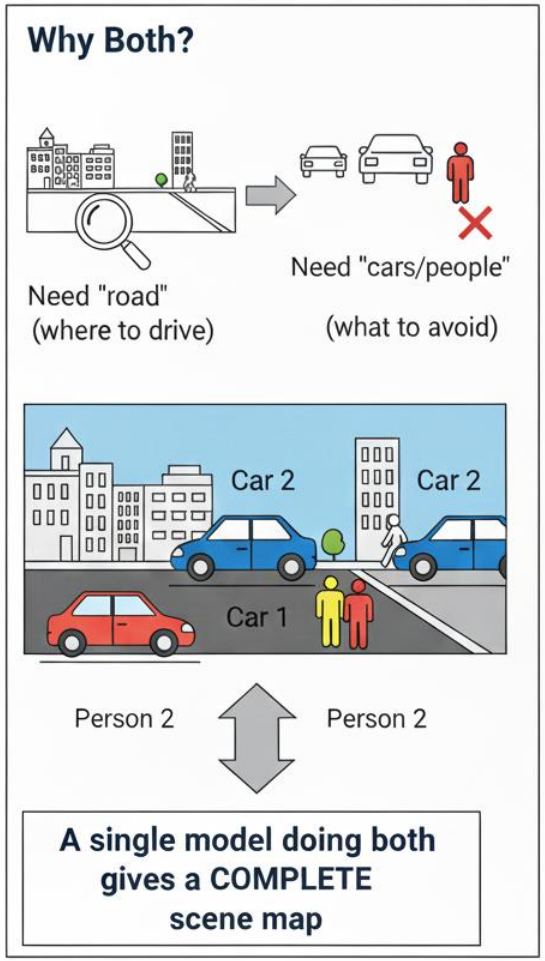
Why Both?

- Example: **Self-driving cars**
- Need “road” → where to drive
- Need “cars/people” → what to avoid
  A single model doing both gives a complete scene map.

Panoptic Quality (PQ)

- Measures both segmentation accuracy + detection quality.
- Formula (simple version):
  - High PQ = segments look good **and** all instances detected
  - PQ = SQ × RQ
    - **SQ:** How cleanly shapes are segmented
    - **RQ:** How many segments were correctly found

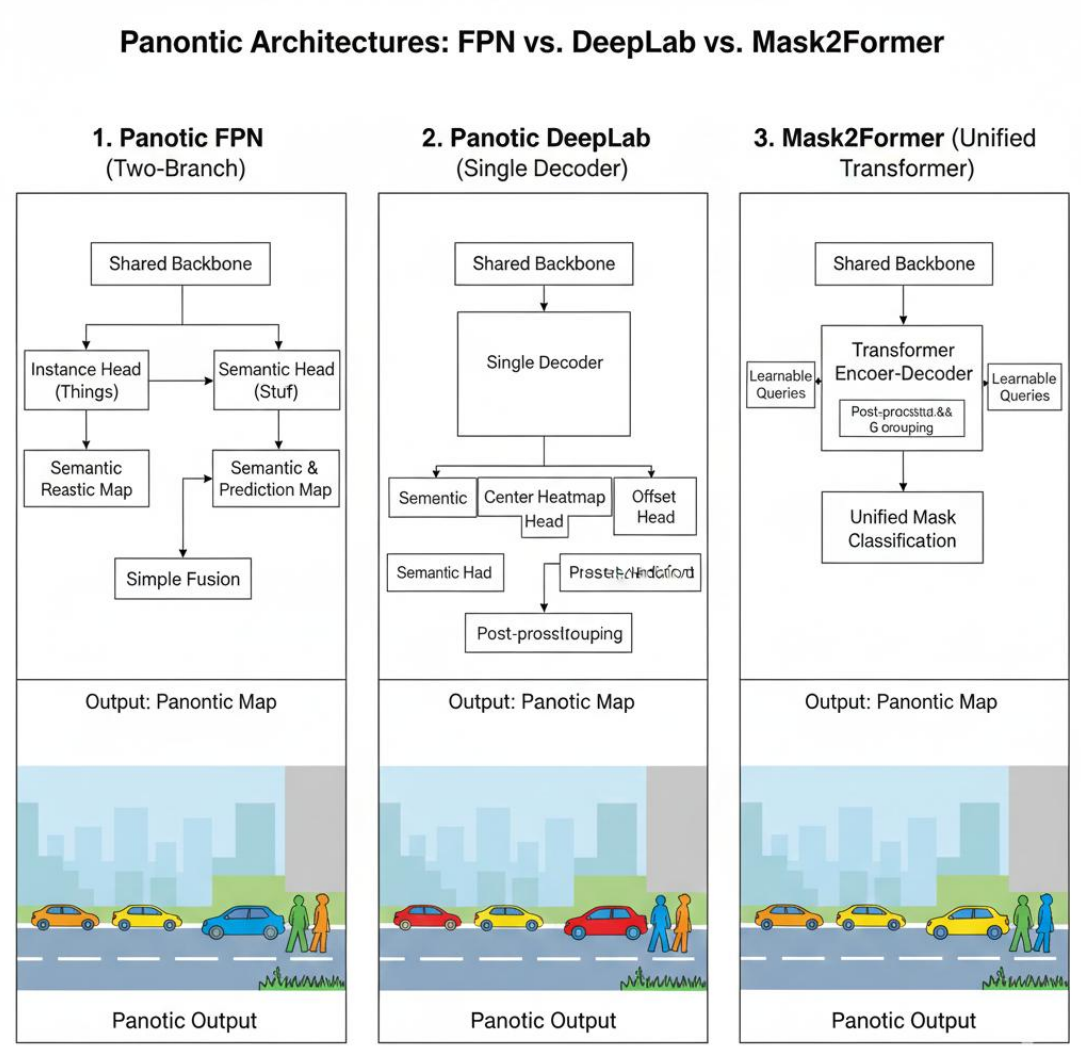




Panoptic Segmentation – Full Scene Understanding

Main Architectures

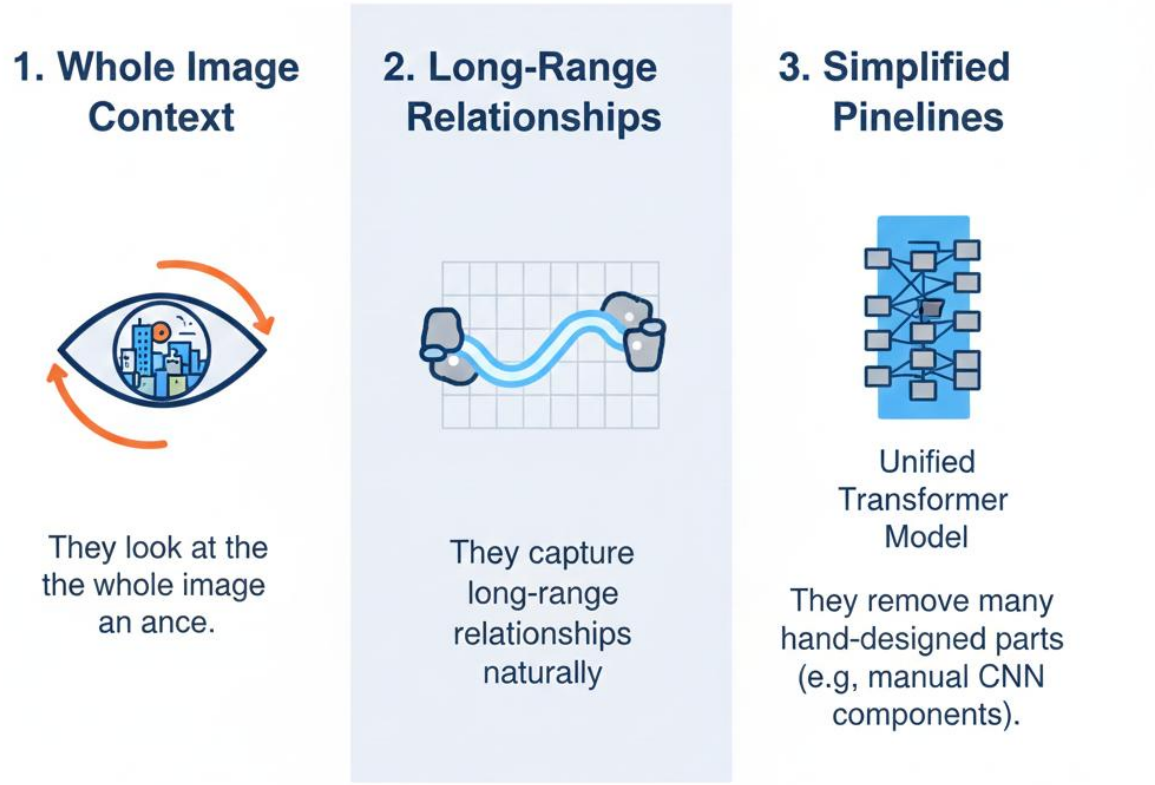
- **Panoptic FPN (Facebook, 2019)**
  - Mask R-CNN branch (instances)
  - FCN branch (stuff)
  - Combine outputs
- **Panoptic DeepLab (Google, 2020)**
  - Single decoder
  - Predicts: semantic map, center heatmap, offsets
  - Groups pixels into instances
- **MaskFormer / Mask2Former (2021/2022)**
  - Transformer-based
  - One unified mask classification approach
  - Works for semantic, instance, and panoptic



Transformers for Segmentation - DETR, SegFormer, SAM

Why Transformers Matter in Vision

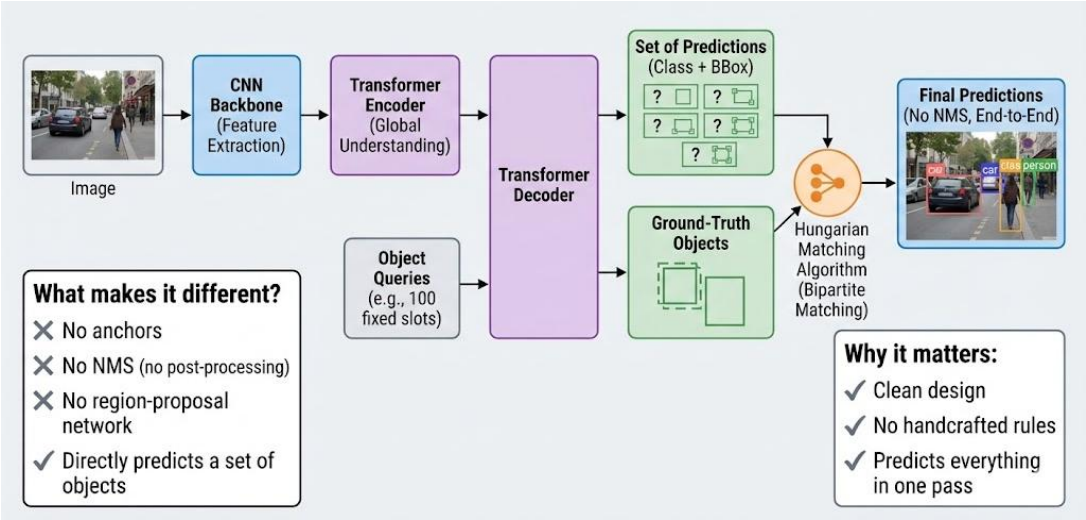
- Transformers changed NLP first (BERT, GPT).
  Now they dominate vision because:
  - ❖ They look at the whole image at once
  - ❖ They capture long-range relationships naturally
  - ❖ They remove many hand-designed parts used in older CNN pipelines



Transformers for Segmentation - DETR, SegFormer, SAM

DETR (2020) — End-to-End Object Detection

- **What makes it different?**
  - No anchors
  - No NMS (no post-processing)
  - No region-proposal network
  - Directly predicts a **set** of objects



- **How it works (simple idea):**
  - CNN extracts image features
  - Transformer encoder adds global understanding
  - Transformer decoder uses **object queries** (100 fixed slots)
  - Each query tries to represent one object

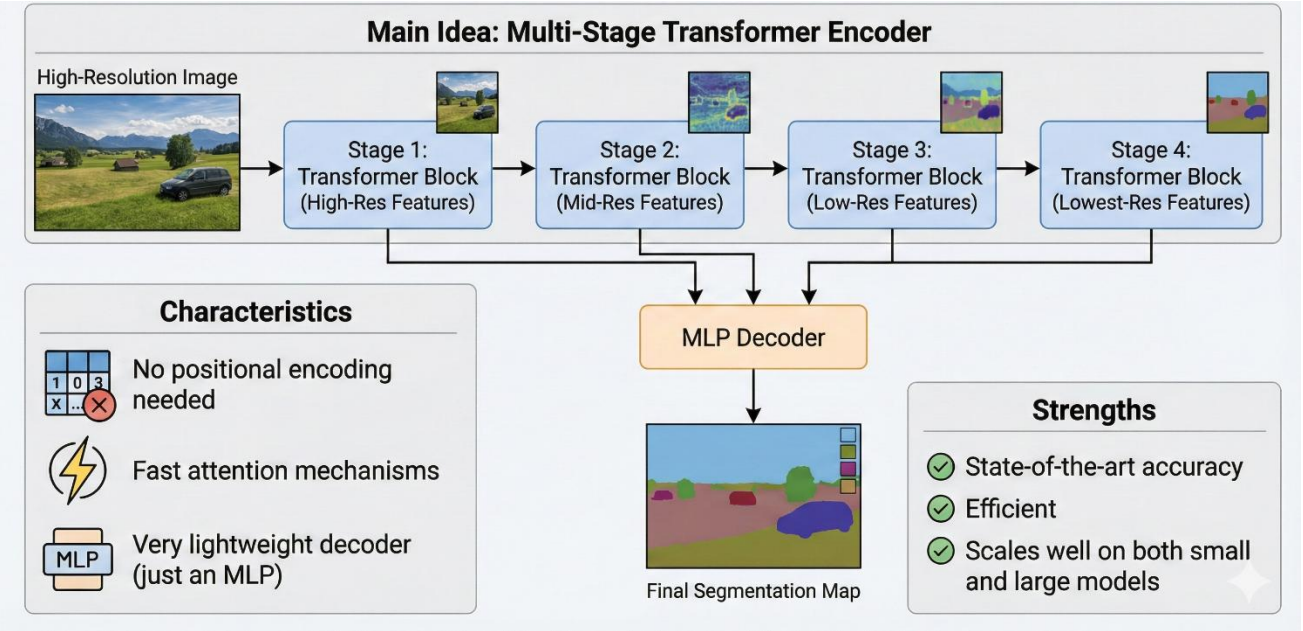
- **Matching Step:**
  - A Hungarian algorithm pairs predictions with ground-truth objects.

- **Why it matters:**
  - Clean design
  - No handcrafted rules
  - Predicts everything in one pass

Transformers for Segmentation - DETR, SegFormer, SAM

SegFormer (2021) — Efficient & Clean Semantic Segmentation

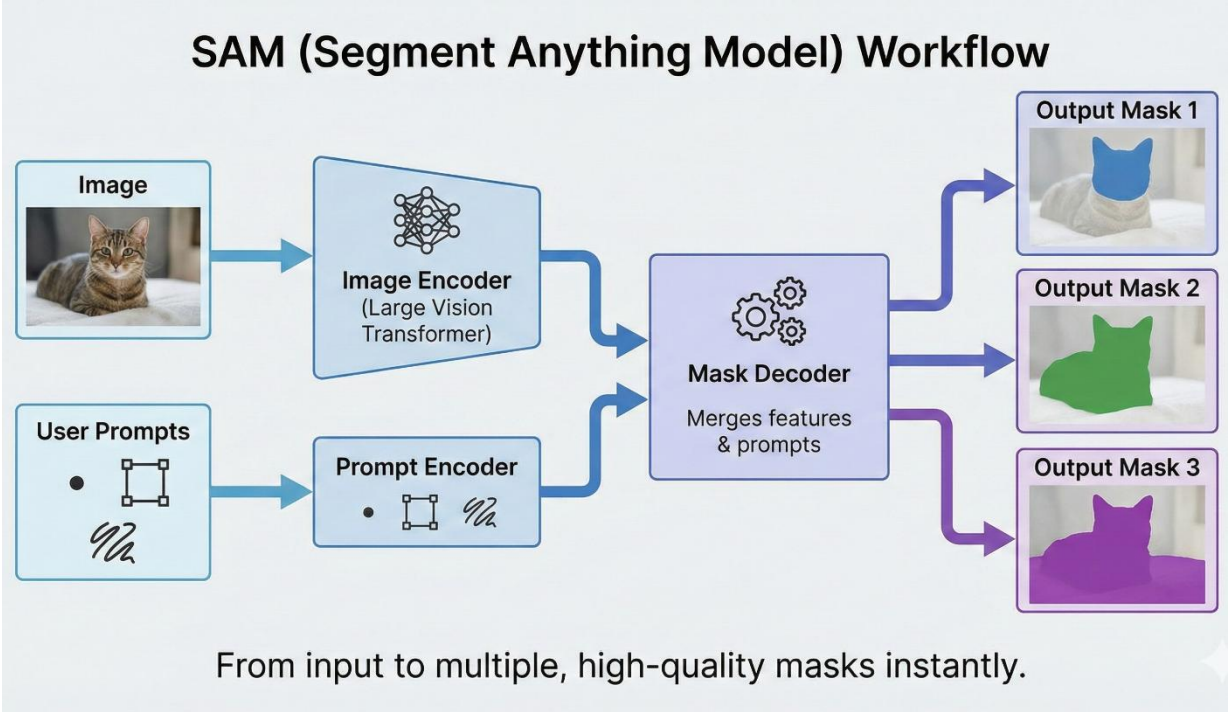
- **Main idea:**
  - A transformer encoder that works in **multiple stages** (from high-resolution to low-resolution).
- **Characteristics:**
  - No positional encoding needed
  - Fast attention mechanisms
  - Very lightweight decoder (just an MLP)
  - Combines features from all stages for strong segmentation
- **Strengths:**
  - State-of-the-art accuracy
  - Efficient
  - Scales well on both small and large models



Transformers for Segmentation - DETR, SegFormer, SAM

SAM – Segment Anything Model (2023)

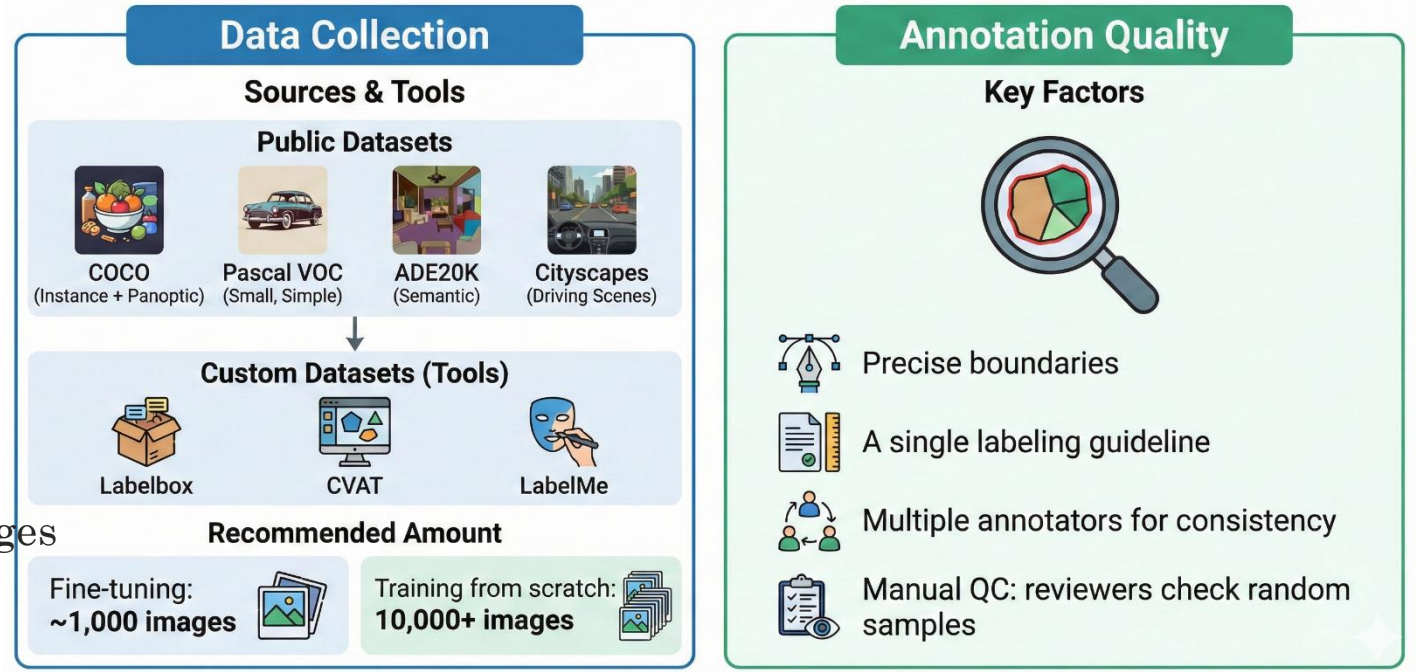
- **What makes SAM special?**
  - Trained on the **largest segmentation dataset ever** (1B masks)
  - Works on **any image** without training
  - Takes **prompts** (point, box, mask)
  - Predicts multiple high-quality masks instantly
- **How it works (simple flow):**
  - A large Vision Transformer encodes the image
  - A separate encoder processes the user's prompt
  - A mask decoder merges both and outputs several mask options
- **Why it's a breakthrough:**
  - Zero-shot segmentation
  - Works interactively
  - Powerful for medical, robotics, video, AR/VR



Training Deep Segmentation Models — Practical Guide

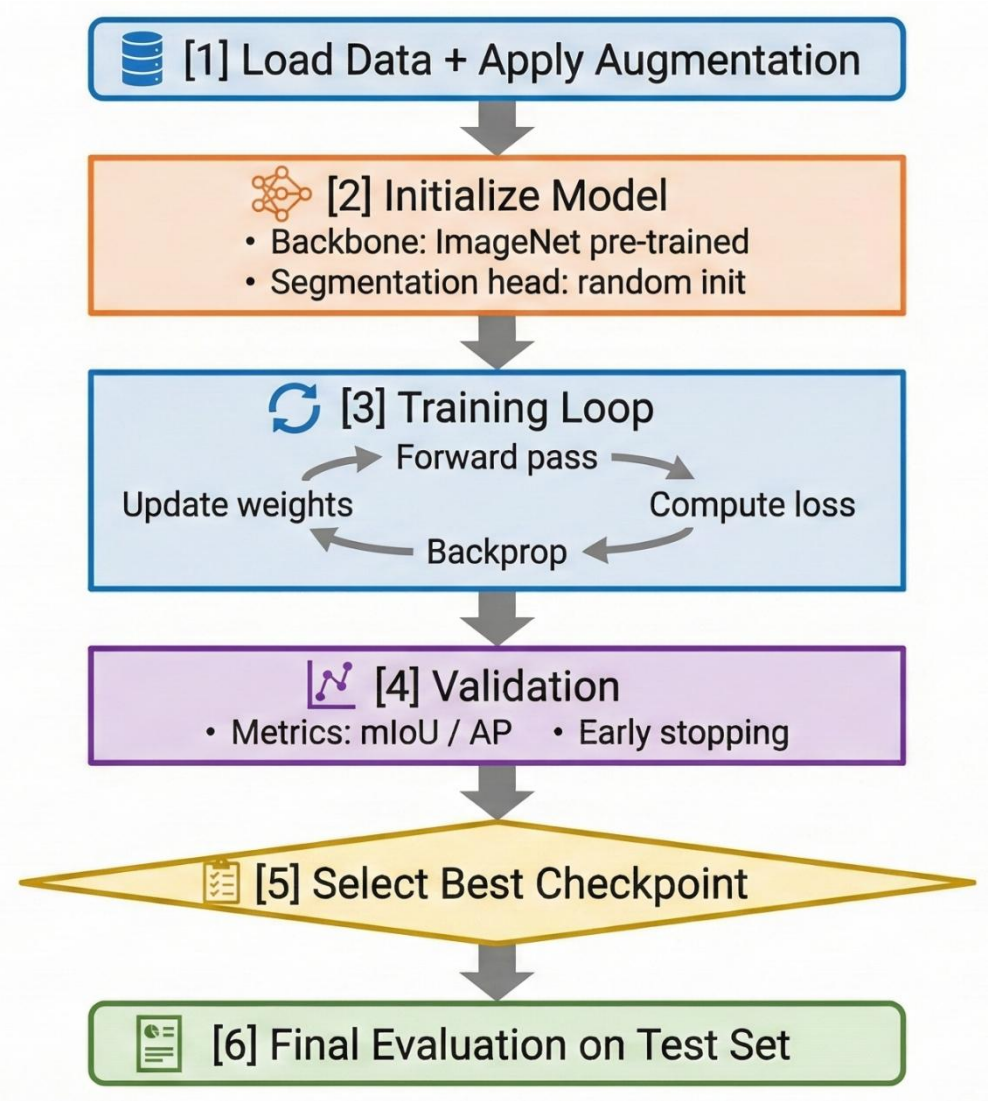
Dataset Preparation

- **Data Collection**
  - Use public datasets or build your own:
    - **COCO** – instance + panoptic
    - **Pascal VOC** – small, simple
    - **ADE20K** – semantic segmentation
    - **Cityscapes** – driving scenes
  - For custom datasets → use **Labelbox, CVAT, LabelMe**.
- **Recommended amount:**
  - **Fine-tuning:** ~1,000 images
  - **Training from scratch:** 10,000+ images
- **Annotation Quality**
  - Segmentation quality depends heavily on labels.
    - Precise boundaries
    - A single labeling guideline
    - Multiple annotators for consistency
    - Manual QC: reviewers check random samples



Training Deep Segmentation Models — Practical Guide

Training Pipeline



Training Deep Segmentation Models — Practical Guide

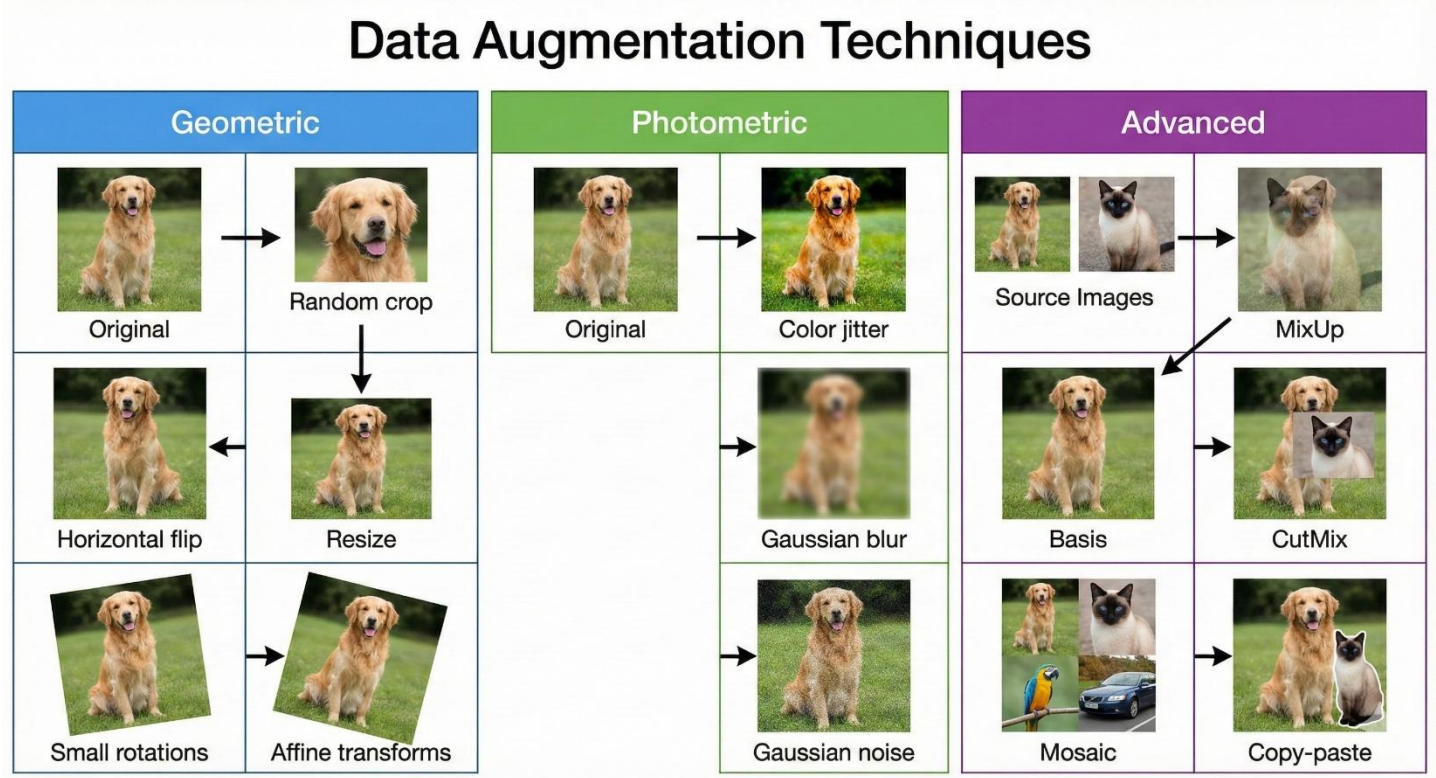
Critical Hyperparameters

Parameter	Typical Value	Notes
Learning Rate	1e-4 → 1e-2	Use LR finder
Batch Size	8–32	Depends on GPU memory
Image Size	512–1024	Larger = better quality
Epochs	50–300	Use early stopping
Optimizer	AdamW, SGD	AdamW for Transformers
LR Schedule	Cosine, Step, Warmup	Warmup = smoother start
Weight Decay	1e-4 → 1e-2	Regularization

Training Deep Segmentation Models — Practical Guide

Data Augmentation

- **Geometric**
  - Random crop
  - Resize
  - Horizontal flip
  - Small rotations
  - Affine transforms
- **Photometric**
  - Color jitter (brightness/contrast/saturation)
  - Gaussian blur
  - Gaussian noise
- **Advanced**
  - MixUp / CutMix
  - Mosaic (YOLO-style)
  - Copy-paste (instance segmentation)



Training Deep Segmentation Models — Practical Guide

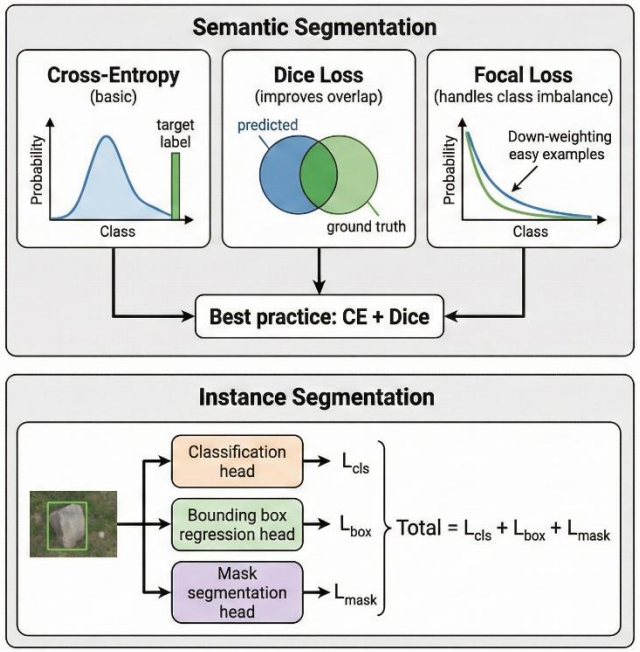
Loss Functions

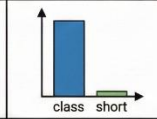
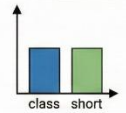
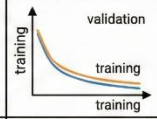
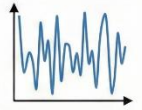
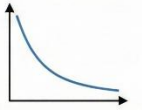
- **Semantic Segmentation**
  - **Cross-Entropy** – basic
  - **Dice Loss** – improves overlap
  - **Focal Loss** – handles class imbalance
- **Best practice: CE + Dice**

- **Instance Segmentation**
  - Standard combined loss:
  - Total = L_cls + L_box + L_mask

Common Pitfalls & Fixes

- **Class imbalance** → weighted loss, oversampling
- **Overfitting** → stronger augmentation, weight decay
- **Small objects missed** → multi-scale training
- **Boundary errors** → boundary-weighted loss
- **LR too high** → unstable training



	Pitfall	Visual Example	Fix
1	Class imbalance		 weighted loss, oversampling
2	Overfitting		 stronger augmentation, weight decay
3	Small objects missed		 multi-scale training
4	Boundary errors		 boundary-weighted loss
5	LR too high		 unstable training

Segmentation Datasets – Benchmarks

Pascal VOC 2012

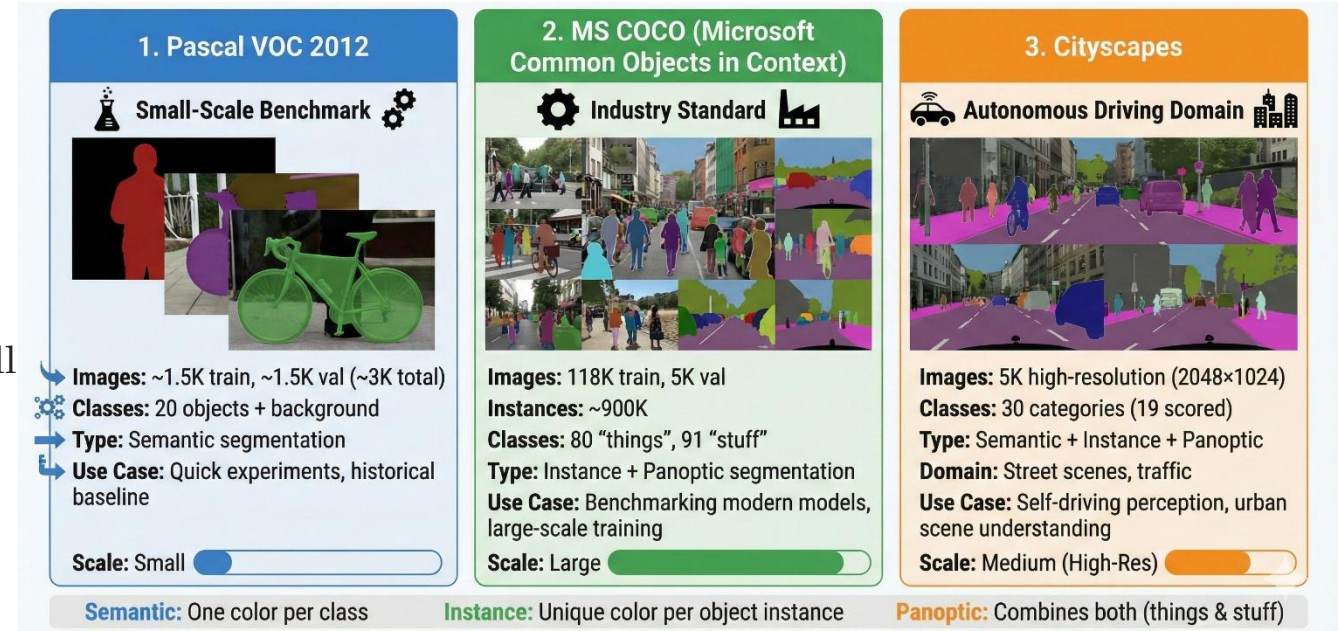
- **Images:** ~1.5K train, ~1.5K val
- **Classes:** 20 objects + background
- **Type:** Semantic segmentation
- **Use Case:** Small-scale benchmark, quick experiments

MS COCO

- **Images:** 118K train, 5K val
- **Instances:** ~900K
- **Classes:** 80 “things”, 91 “stuff”
- **Type:** Instance + panoptic segmentation
- **Use Case:** Industry standard; used for almost all modern models

Cityscapes

- **Images:** 5K high-resolution (2048×1024)
- **Classes:** 30 categories (19 scored)
- **Type:** Semantic + instance + panoptic
- **Domain:** Autonomous driving
- **Use Case:** Street scenes, traffic, self-driving perception



Segmentation Datasets – Benchmarks

ADE20K

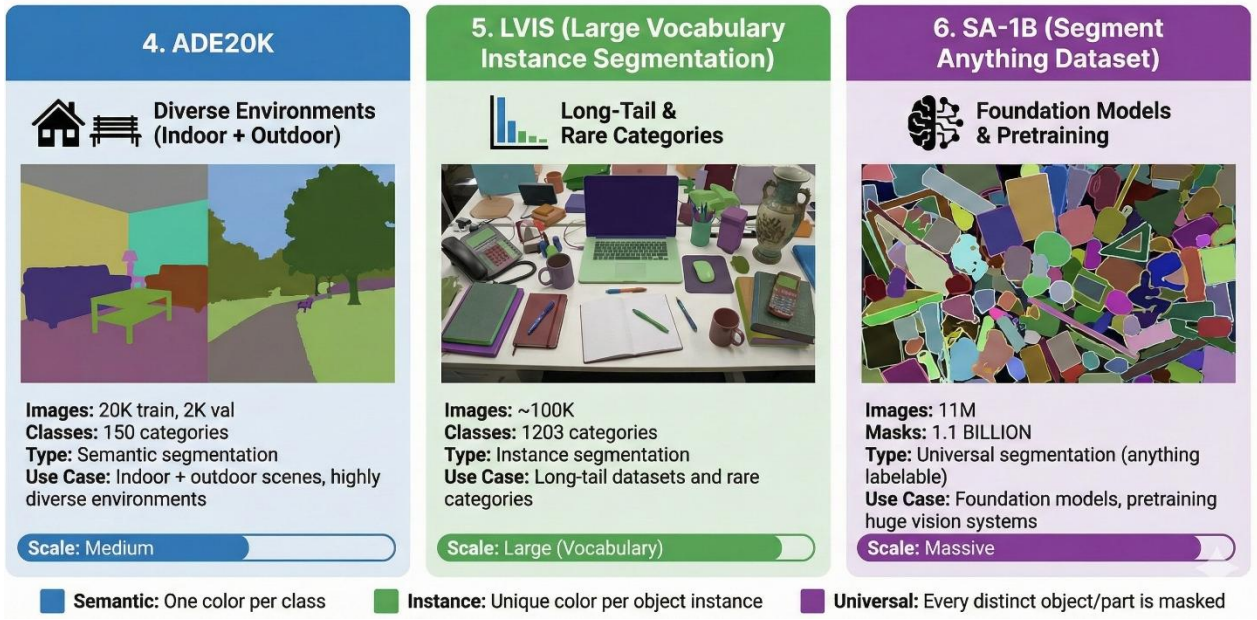
- **Images:** 20K train, 2K val
- **Classes:** 150 categories
- **Type:** Semantic segmentation
- **Use Case:** Indoor + outdoor scenes, highly diverse environments

LVIS (Large Vocabulary)

- **Images:** ~100K
- **Classes:** 1203 categories
- **Type:** Instance segmentation
- **Use Case:** Long-tail datasets and rare categories

SA-1B (Segment Anything Dataset)

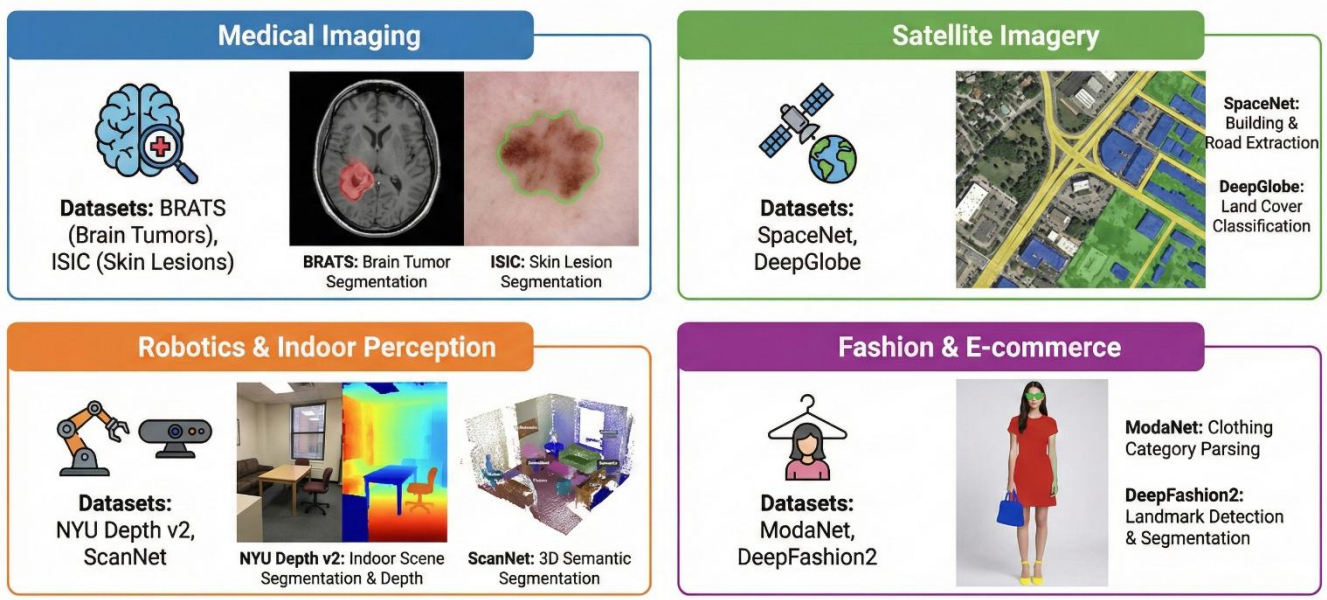
- **Images:** 11M
- **Masks:** 1.1 billion
- **Type:** Universal segmentation (anything labelable)
- **Use Case:** Foundation models, pretraining huge vision systems



Segmentation Datasets – Benchmarks

Domain-Specific Datasets (Examples)

- **Medical:** BRATS (brain tumors), ISIC (skin lesions)
- **Satellite:** SpaceNet, DeepGlobe
- **Robotics:** NYU Depth v2, ScanNet
- **Fashion:** ModaNet, DeepFashion2



These datasets provide specialized data for training segmentation models in specific application domains, enabling precise analysis and automation.

Dataset Comparison Table

Dataset	Size	#Classes	Resolution	Annotation Quality
Pascal VOC	Small	21	Medium	★★★★★
COCO	Large	80/91	Medium	★★★★☆
Cityscapes	Medium	30	High	★★★★★
ADE20K	Medium	150	Mixed	★★★★☆
LVIS	Large	1203	Medium	★★★★☆

Thank You

For Your Attention